Combating Noise Pollution: Policy, Exposure, and Welfare*

Takeru Sugasawa HRAFJ
Shinsaku Takikawa LSE
Takanao Tanaka UC Berkeley

November 2026

Link to the most recent draft

This paper provides the first evidence on how concerted national efforts to combat noise pollution affect noise exposure and population welfare. We assemble two decades of largely untapped nationwide measurements of road-traffic noise from over 14,000 locations across Japan and detailed data on noise-mitigation policies obtained through freedom-of-information requests. We present four findings. First, from 2002 to 2022, roadside noise fell by 2.8 decibels, equivalent to roughly a 20% decline in perceived loudness. Second, about 75% of this decline is attributable to noise-control policies, including tighter vehicle noise-emission standards and the expansion of low-noise pavement. Third, residents are willing to pay 1.42% of residential land value for a one-decibel reduction in road-traffic noise. Fourth, Japan's national noise-control policies generated approximately 6.8 trillion yen (US$57 billion) in benefits over two decades. These benefits suggest substantial unrealized gains in countries with weaker noise-control policies, such as the United States.

Keywords: Noise pollution; environmental regulation; policy evaluation; nonmarket valuation

JEL Codes: Q53, Q58, R31, R41

*We thank Michael Anderson, Sefi Roth, James Sallee, and Joseph Shapiro, as well as seminar and conference participants at Applied Regional Economics Workshop, IPWSD (SIPA), JTREE, LSE, Society for Environmental Economics and Policy Studies, Sophia University, and UC Berkeley for insightful and constructive comments. We acknowledge financial support from the Institute of Economic Research at Hitotsubashi University. All errors are our own. Emails: Sugasawa, sugasawa@hrf.or.jp; Takikawa, jamubin@gmail.com; Tanaka, takanao.tanaka.0323@gmail.com.

1 Introduction

Noise pollution—unwanted or harmful sound—is more than a nuisance. It disrupts sleep, impairs cognition, and is linked to poorer health and lower productivity (WHO, 2018).[1] The European Commission recognizes it as Europe's second-largest environmental health threat, behind only air pollution (EC, 2023). Despite this recognition, an estimated 110 million Europeans and 47 million Americans remain exposed to harmful transportation noise today (EEA, 2025; Shkembi et al., 2026). How should governments tackle this widespread environmental threat?

Governments around the world take starkly different approaches to noise pollution. The United States, for example, enacted the Noise Control Act in 1972 and the Quiet Communities Act in 1978 (U.S. Congress, 1972, 1978). Yet in 1982, the Environmental Protection Agency (EPA) defunded these laws and left much of the responsibility for noise control to state and local governments (EPA, 2026). This fragmented approach has drawn sustained criticism across scientific, legal, and policy communities. For nearly half a century, residents in many jurisdictions have remained, in the American Public Health Association's words, "without legal recourse to stop harmful noise exposures" (APHA, 2021).

Japan offers a striking contrast: since the mid-1990s, it has mounted a concerted national effort to combat noise pollution. In 1995, the Supreme Court upheld the Japanese government's liability for road-traffic-noise damages in the landmark Route 43 case, finding that the harm exceeded "what residents could reasonably be expected to tolerate" (Supreme Court of Japan, 1995). Within months, five government ministries declared that they "recognized the gravity" of the ruling and pledged to coordinate nationwide efforts to combat road-traffic noise (Interministerial Council, 1995). By the mid-2000s, the national government was allocating roughly ¥40 billion (US$333 million) annually to noise-mitigation policies. Three policy measures were central to this effort: nationwide monitoring of road-traffic noise, investment in noise-reducing infrastructure, and tighter vehicle noise-emission standards.

Can concerted national efforts to combat noise pollution reduce noise exposure and improve population welfare? We provide the first evidence. We study Japan's efforts to control road-traffic noise, a major source of noise pollution, from 2002 to 2022—a period of sustained national action following the landmark 1995 Supreme Court ruling.[2]

We assemble what is, to our knowledge, the most comprehensive longitudinal dataset of road-traffic noise measurements to date, drawing on largely untapped administrative records

[1] See Guski et al. (2017); Basner and McGuire (2018) for evidence on annoyance and sleep disturbance, van Kempen et al. (2018); Greenhill (2026) for health effects, and Dean (2024) for cognition and productivity.

[2] In Europe, an estimated 92 million people are exposed to harmful levels of road-traffic noise, compared with 18 million from rail and 2.6 million from aircraft (EEA, 2025).

from Japan's Continuous Monitoring of Road Traffic Noise (CMRTN) system. The Ministry of the Environment established CMRTN as a central pillar of Japan's road-traffic noise policy to assess both "the current state of road traffic noise and the effects of countermeasures." The system therefore aims to collect data on residents' noise exposure in a "nationwide, continuous, and consistent" manner (MOE, 2022). Our harmonized dataset contains 46,544 measurements across 14,261 road segments from 2002 to 2022—information that previous research has scarcely exploited despite its extensive coverage. We link these measurements to data on noise-mitigation infrastructure from the Road Environmental Census, which we obtained through a Freedom of Information Act request. The census records low-noise pavement, noise barriers, roadside buffer strips, and sound-absorbing panels on elevated structures from 2000 to 2010.

This paper answers four empirical questions. First, how did road-traffic noise change in Japan over the two decades from 2002 to 2022 and who benefited from the change? Second, to what extent did Japan's national noise-control policies—particularly tighter vehicle sound-emission standards and expanded noise-mitigation infrastructure—contribute to these changes? Third, what is residents' willingness to pay for quieter roads? Finally, what are the monetized benefits of Japan's noise-mitigation policies?

Our answers to all four questions are striking. First, road-traffic noise in Japan fell markedly from 2002 to 2022. Noise levels declined by 2.8 decibels over this period. Because the decibel scale is logarithmic, this decline corresponds to nearly a 50 percent reduction in sound intensity and roughly a 20 percent reduction in perceived loudness. To put this decline in context, it is comparable to halving traffic volume, lowering the speed limit from 50 to 30 km/h (FOEN, 2025), switching a robot vacuum from standard to quiet mode (Hales, 2024), or moving from station ticket gates to a coffee shop (Sueoka et al., 2009). At the same time, the probability that a road exceeded environmental noise standards declined by about 30 percentage points, or nearly 50%. Despite the overall improvement, the pace of noise reduction slowed over time; about two-thirds of the decline occurred between 2002 and 2012, the first half of our study period. Initially noisier locations experienced faster declines, indicating convergence in noise exposure across locations. We find no evidence of larger reductions in areas with a higher share of college graduates.

Second, Japan's noise-control policies account for about 75% of the observed decline in noise. Specifically, Japan tightened vehicle noise-emission standards from the late 1990s onward, and these changes account for about 28% of the decline. The rapid expansion of low-noise pavement accounts for 35%, and the installation of noise barriers for 8.3%. We estimate these contributions by constructing an acoustic energy prediction model that combines engineering estimates of each intervention's effect with observed changes in its prevalence. Low-noise pavement, for example, expanded from 28.0% to an estimated 68.1%

of measurement sites between 2002 and 2022, while noise barriers expanded from 8.3% to an estimated 13.0%. For vehicle standards, we combine changes in permissible noise limits with the share of vehicles subject to the new standards and fleet turnover. The model also accounts for changes in traffic volume, heavy-vehicle shares, and electric and hybrid vehicle adoption, which together explain only about a quarter of the observed noise reduction.

Third, residents living near roads are willing to pay 1.42% of residential land value for a 1 dB reduction in road-traffic noise. We estimate this willingness to pay using a hedonic design that exploits the replacement of conventional dense-graded asphalt with low-noise pavement, Japan's most widely adopted noise-mitigation intervention. Low-noise pavement reduces noise by about 2.55 dB and raises residential land prices within 100 m of the road by about 3.6%, implying that a 1 dB reduction in noise is capitalized into a 1.42% increase in land prices. Beyond 100 m, low-noise pavement has little detectable effect on residential land prices, likely because the associated noise reductions become less perceptible.

Finally, Japan's noise-control policies between 2002 and 2022 generated approximately 6.8 trillion Japanese yen (US$57 billion) in monetized benefits. For the roughly 15.5 million households living within 100 m of major roads, these benefits amount to about 438,000 yen (US$3,700) per household, or 21,900 yen (US$180) per year when spread evenly over 20 years. We obtain this estimate by combining the estimated willingness to pay per decibel with the policy-attributable reduction in noise and the value of affected residential land.

This paper contributes to several strands of literature. First, to our knowledge, this study provides the first ex post nationwide estimates of the welfare benefits of road-noise control policies worldwide. In the United States, benefit–cost analysis has long played a central role in evaluating environmental policies.[3] The U.S. EPA estimates that the benefits of the 1990 Clean Air Act Amendments exceed their costs by more than 30 to one (U.S. EPA, 2011), while those of its 2024 Lead and Copper Rule Improvements could exceed costs by up to 13 to one (U.S. EPA, 2024). For noise pollution, the U.S. EPA examined the benefits and costs of transportation-noise abatement in 1977, but these estimates were prospective rather than based on the realized effects of implemented policies (National Research Council, 1977). Since then, comparable nationwide federal benefit–cost evaluations of noise-control policies have been largely absent, presumably because few federal noise-control policies are available for assessment.[4] The European Commission (2023), in contrast, recognizes noise pollution

[3]Researchers have also extensively studied the economic consequences of federal laws to reduce air pollution (Greenstone, 2002; Walker, 2013), clean up rivers and lakes (Keiser and Shapiro, 2019), improve drinking-water safety (Keiser et al., 2026), clean up hazardous waste sites (Greenstone and Gallagher, 2008), and protect biodiversity (Frank et al., 2025).

[4]Non-federal research institutions have also examined the potential benefits and costs of noise mitigation. For example, the National Academy of Engineering (2010), a private nonprofit institution, reviewed existing economic evidence on highway-noise control and called for further benefit–cost analyses, rather than

as the second-largest environmental health threat after air pollution and conducts cost–benefit analyses of noise-control measures. It estimates that these measures could generate 10 euro in societal benefits for every 1 euro spent. Yet this analysis focuses on simulating policy scenarios rather than estimating the benefits and costs of implemented noise-control policies.

We also provide the first nationwide ex post estimates of the welfare benefits of environmental policy in Japan using micro-level data. Even more than half a century after Japan introduced regulations on air, water, and noise pollution and established the Environment Agency, the nationwide economic benefits of these efforts remain largely unknown (Botta and Yamasaki, 2020). One possible reason is a reluctance to place a monetary value on the environment. In the 1970s, after pollution had caused severe illness and loss of life, efforts to weigh the costs of environmental protection against human health and life were "subject to strong criticism and condemnation" (Matsuno and Ueta, 2002, pp. 145–146). The government's 1973 Environmental White Paper echoed this view: "A clean environment has great value that *cannot be expressed in monetary terms*." It continued: "Many aspects of it must be protected, *no matter how great the cost to society*" (authors' translation; Environment Agency of Japan, 1973). This historical reluctance may help explain why nationwide estimates of the welfare benefits of environmental policies, including those targeting noise pollution, remain scarce in Japan.[5]

Our study is also the first to document nationwide long-run changes in noise pollution and assess the contributions of different noise-mitigation interventions. Despite extensive evidence on the adverse effects of noise exposure on health, cognition, and well-being, basic facts about noise pollution—how it has evolved over time and who has benefited from these changes—remain scarce. This gap partly reflects data limitations: the United States and Europe largely rely on modeled measures of noise exposure (BTS, 2022; EEA, 2025), providing limited direct evidence on changes in measured noise levels.[6] China conducts extensive nationwide noise monitoring, but site-level longitudinal data are not publicly available.[7] We

evaluating the realized effects of implemented policies. More recently, the U.S. Department of Transportation acknowledged that it lacks a recommended methodology for monetizing noise reductions and generally recommends treating these benefits qualitatively in benefit–cost analyses (U.S. DOT, 2025).

[5]Kochi et al. (2002) estimate the benefits and costs of Japan's air-pollution controls using a modeling approach, without micro-level data or a rigorous empirical design. Kang et al. (2024) and Nishitateno and Burke (2021) use micro-level data to estimate regional benefits in the Tokyo metropolitan area and in Aichi and Mie prefectures, respectively.

[6]Comparisons across noise-mapping rounds require caution. The U.S. Bureau of Transportation Statistics notes that its 2014 prototype is not directly comparable with later releases and that changes in traffic-data reporting can generate apparent changes in noise (BTS, 2022). In Europe, methodological changes made the original 2017 and 2022 estimates non-comparable, so EEA trend assessments reconstruct the 2017 baseline using the 2022 methodology (Blanes et al., 2023, pp. 6, 11).

[7]Existing studies use measurements from only a few cities or aggregate monitoring data to the city level.

use two decades of noise measurements to document long-run trends and data on different types of noise-mitigation infrastructure to assess their contributions to these trends.

Finally, we contribute to the literature on households' willingness to pay for quieter environments by using directly measured roadside noise. Existing studies exploit variation in flight paths, airport operations, transportation infrastructure, and noise-mitigation interventions to estimate how noise is capitalized into property values (Boes and Nüesch, 2011; Winke, 2017; Ahlfeldt et al., 2019; von Graevenitz, 2018; Allroggen et al., 2025; Sugasawa et al., 2025; Lindgren, 2021; Moretti and Wheeler, 2025). These studies generally rely on modeled noise exposure to infer willingness to pay for a 1 dB reduction in noise. We estimate the effects of low-noise pavement on both measured roadside noise and residential land values, allowing us to estimate willingness to pay per decibel of observed noise reduction. Because low-noise pavement is widely adopted, our estimates are directly relevant for evaluating noise control at scale.

2 Background

2.1 Road-Traffic Noise: Sources, Impacts, and Mitigation

This section provides background on road-traffic noise, its sources, and policies to mitigate noise pollution.

Noise pollution imposes costs on households by disrupting daily life and harming health and cognition. Noise causes annoyance and disrupts sleep (WHO, 2018; Guski et al., 2017; Basner and McGuire, 2018). Research also documents adverse health effects. Long-term road-traffic noise exposure may increase the risk of ischemic heart disease (van Kempen et al., 2018), while prenatal noise exposure can harm infant health (Greenhill, 2026). Beyond these health effects, noise can impair cognitive performance and reduce productivity (Dean, 2024). These disruptions and risks may motivate households to pay for quieter homes.

Road traffic is the most widespread source of noise pollution (WHO, 2018). Across Europe, 92 million people faced elevated road-traffic noise in 2022, far exceeding the 18 million exposed to railway noise and 2.6 million exposed to aircraft noise (EEA, 2025). Noise exposure was also widespread in the United States, where road noise affected 74.7 million people in 2018 (BTS, 2020).[8]

For example, Han et al. (2018) use cross-sectional measurements from Shenzhen, while Xu et al. (2020) analyze city-level data for 111 cities over 1991–2017.

[8]The European figures use a threshold of 55 dB L_{den}, which includes penalties for evening and nighttime noise. The U.S. figures use 45 dB $L_{\text{Aeq},24h}$, a 24-hour energy average without these penalties. Differences in thresholds and measures prevent direct comparison of the exposure counts.

Vehicle noise comes from three main sources: propulsion, tire–road interaction, and airflow. First, engines, motors, and exhaust systems produce propulsion noise, especially during acceleration. Second, tire–road interaction generates noise through tire vibration and the compression and release of air at the contact surface. The pavement's surface texture affects the level of this noise. For example, conventional dense-graded asphalt can generate more tire–road noise because air trapped beneath the tires cannot escape easily. Lastly, turbulence around a moving vehicle generates aerodynamic noise, which becomes more prominent at high speeds (Rasmussen et al., 2007).

Policymakers use a range of measures to reduce road-traffic noise, either by reducing noise at its source or by limiting its propagation. Vehicle-noise standards aim to reduce sound emissions from individual vehicles by setting maximum permissible noise levels. To meet these limits, automakers can install quieter mufflers, add sound insulation around the engine, and use components that dampen vibration. Additionally, low-noise pavements, such as porous asphalt, reduce tire–road noise through their texture and porous structure. Traffic-management measures can further reduce noise by lowering traffic volumes or vehicle speeds. To limit noise propagation, policymakers can install noise barriers and absorptive panels, which block or absorb sound before it reaches nearby homes (Brown and van Kamp, 2017).

Governments around the world take markedly different approaches to noise pollution. The WHO provides noise-exposure guidelines to protect population health, but individual countries decide whether and how to translate them into policy (WHO, 2018). For example, the United States enacted the Noise Control Act in 1972 and the Quiet Communities Act in 1978 (U.S. Congress, 1972, 1978). However, in 1982, these laws became "essentially unfunded," and there has been "little federal action on noise" for nearly half a century (EPA, 2026; APHA, 2021). Much of the practical responsibility for noise control now rests with state and local governments. In contrast, the EU's Environmental Noise Directive establishes a coordinated framework for managing noise pollution. It requires member states to prepare noise maps and action plans for major urban areas and transport infrastructure, in principle every five years, while leaving the choice and implementation of specific measures largely to national and local authorities (EU, 2002).

Evidence on how noise-mitigation measures have changed road-traffic noise over long periods remains limited primarily due to the lack of long-term, consistent, and directly measured noise data. The U.S. National Transportation Noise Map provides nationwide modeled estimates of average transportation noise, but it does not provide repeated, directly measured noise levels at comparable roadside locations over time (BTS, 2022). The EU has conducted repeated strategic noise mapping since 2007, but methodological differences

across countries and reporting rounds limit long-run comparability.[9]

2.2 Noise-Mitigation Policies in Japan

Japan provides a unique setting for studying noise-mitigation policies because it has unusually rich and systematic data on road-traffic noise and mitigation measures.

Japan began combating noise pollution in the nineteenth century. As early as 1878, Japan introduced what historians identify as its first explicit noise regulation, prohibiting singing, music, and other noisy activities after midnight when they disturbed others' sleep. With the spread of motor vehicles, Japan extended regulation to road transportation in 1919, requiring vehicles not to generate excessive noise. In 1968, Japan enacted the Noise Regulation Act, a landmark national law addressing noise from factories, construction, and motor vehicles. The act has provided the foundation for Japan's noise-control policies for more than half a century, and local governments play a central role in implementing mitigation measures (Government of Japan, 1968; MOE, ndb). Environmental quality standards complement these efforts by setting targets of 70 dB during the daytime and 65 dB at night for areas adjacent to major arterial roads (MOE, 1998).

Our extensive interviews with officials from multiple local and national government agencies inform our interpretation that the Route 43 litigation served as a catalyst for stronger action against road-traffic noise. In 1976, residents along Route 43 and the Hanshin Expressway brought suit over severe roadside noise. Nineteen years later, in July 1995, the Supreme Court upheld government liability and an award of damages to affected residents—the Court's first road-traffic-noise ruling to hold road administrators liable (Supreme Court of Japan, 1995). The ruling prompted a rapid policy response. The government later stated that the relevant agencies "took the ruling seriously," and in August, five ministries and agencies—the Environment Agency, National Police Agency, Ministry of Construction, Ministry of Transport, and Ministry of International Trade and Industry—agreed on countermeasures for the Route 43 corridor. By December, they had extended this framework to other areas facing severe road-traffic noise. The litigation thus helped turn a decades-long local dispute into a broader national policy response (Interministerial Council, 1995).

Vehicle noise-emission standards are a key component of Japan's intensified efforts to control road-traffic noise. Japan substantially tightened vehicle noise-emission standards

[9] In the 2007, 2012, and 2017 reporting rounds, member states could use national calculation methods, and the European Environment Agency cautions that estimates across countries and years may therefore not be fully comparable. A common methodology, CNOSSOS-EU, became mandatory in 2019 and was first used EU-wide in the 2022 reporting round. The EEA consequently constructed a harmonized 2017 baseline to enable comparisons with 2022.

between 1998 and 2001, targeting noise directly at its source.[10] A central component sets a maximum noise level for new vehicles under a standardized acceleration test, when vehicles tend to be particularly loud.[11] Automakers must ensure that new vehicles comply with these standards before bringing them to market. The reforms lowered this ceiling by roughly 2 dB for passenger cars and lighter vehicles and 2–3 dB for larger vehicles, though the exact figure depends on vehicle type and size. In practice, the effects of the reforms on roadside noise may be smaller than the reductions in regulatory limits. First, the limits apply to acceleration tests under controlled conditions, rather than directly to measured roadside noise. Second, some vehicles already meet the tighter standards. Third, the standards apply to new vehicles, so their effects emerge gradually as the fleet turns over.

Policymakers also expanded noise-mitigation road infrastructure. Low-noise pavement emerged as the most widely adopted measure on national roads. In particular, porous asphalt emerged in Japan in the late 1990s and spread rapidly thereafter. Between 2000 and 2010, its coverage increased from about 10% to 42%, more than quadrupling over the decade, compared with 4% to 7% for noise barriers, 1.8% to 2.1% for roadside buffer strips, and 0.2% to 0.6% for sound-absorbing panels.[12]

Alongside these noise mitigation measures, nationwide monitoring became another central pillar of Japan's road-traffic noise policy. The Ministry of the Environment recognized the importance of assessing both "the current state of road traffic noise and the effects of countermeasures." The ministry therefore established the Continuous Monitoring of Road Traffic Noise (CMRTN), which tracks residents' exposure along major roads (MOE, 2022). The system collects these data in a "nationwide, continuous, and consistent" manner. National data are consistently available from 2002 through the time of writing, providing an unusually rich basis for studying the long-run evolution and effectiveness of road-traffic noise policy.

3 Data

3.1 Continuous Monitoring of Road Traffic Noise

We obtain road-traffic noise outcomes for 2002–2022 from the Continuous Monitoring of Road Traffic Noise (CMRTN) program of Japan's Ministry of the Environment (MOE) (MOE,

[10] Japan had regulated vehicle noise since the 1950s and introduced acceleration-noise standards in 1971, but the 1998–2001 reforms marked a major new round of tightening after the earlier regulatory cycle.

[11] The test measures exterior noise as a vehicle accelerates under prescribed conditions; the applicable speed, gear, and measurement procedures depend on the vehicle category and regulation.

[12] Roadside buffer strips separate traffic lanes from nearby homes, increasing the distance over which sound travels. Sound-absorbing panels beneath elevated roads absorb reflected traffic noise.

2022). The CMRTN monitors major road corridors that pass through or near residential areas across Japan. It covers expressways, urban expressways, national roads, and prefectural roads with at least two lanes.[13] Monitoring coverage has expanded gradually.[14] Our data cover 46,481 kilometers (approximately 28,882 miles) of monitored road corridors.

The CMRTN monitors noise levels for road segments on a rotating schedule. Each CMRTN segment covers a stretch of road with approximately constant noise levels; the CMRTN splits roads into segments where traffic conditions, road structure, or noise-control infrastructure change, or where roads cross municipal boundaries.[15] We link these observations across years to construct a panel of road segments (see Appendix Section A-1.1). Our dataset contains 14,261 road segments, with an average length of approximately 4.1 kilometers (2.5 miles). Figure 1 illustrates the geographic coverage of the monitoring system. Panels A and B show monitoring locations across Japan and in the Tokyo metropolitan area. In principle, local governments measure noise levels for each road segment every five years.[16] Figure 2 illustrates how local governments rotate monitoring across road segments, using Minato ward from 2018 to 2022 as an example. Panel A combines monitoring locations, target roads, and road segments across 2018–2022, while Panels B–F show each year separately. Appendix Figure A1 presents corresponding maps for all 23 wards of Tokyo, with observations combined over 2018–2022.

Local governments follow MOE guidelines on when, where, and how to measure road-traffic noise, aiming to capture representative noise levels for each road segment. They select measurement locations within each segment so that the measurements reflect its traffic conditions, road structure, noise-control infrastructure, and nearby buildings. At the selected monitoring point, local governments normally measure noise continuously for 24 hours on a day with traffic conditions typical of the year; they avoid days with rain, snow, strong winds, construction, or other temporary disturbances. The MOE also specifies how to separate road-traffic noise from other sounds, where to place microphones, which equipment to use, and how to measure traffic. Our data include information on measurement conditions, such as the distance from the roadway, microphone height, and measurement month. Appendix

[13] Some municipal governments also monitor noise along municipal roads.

[14] Coverage expanded substantially in 2012, when a decentralization reform gave municipalities responsibility for monitoring.

[15] Relevant traffic characteristics include volume, speed, vehicle composition, and daily traffic patterns. Road characteristics include the number of lanes, elevated or depressed structures, low-noise pavement, and noise barriers. The density and arrangement of nearby buildings also matter.

[16] In our data, approximately 90% of consecutive observations for the same segment occur within five years. Local governments choose measurement locations and adjust monitoring schedules to local conditions. Budgets, administrative practices, road changes, and field conditions can affect how often they monitor a segment. They may monitor segments with stable conditions less often than those where roads or traffic conditions change.

Section A-1.2 provides further details on the measurement procedures.

The CMRTN measures noise using equivalent continuous sound levels in decibels (dB), and we combine its daytime and nighttime measurements to construct a 24-hour equivalent noise level. The equivalent continuous sound level summarizes noise that varies over time as a constant level with the same acoustic energy over the same period. Decibels express noise levels on a logarithmic scale. A 10 dB increase corresponds to a tenfold increase in acoustic energy, while a 1 dB increase corresponds to approximately 26% more acoustic energy. The data report daytime (6:00–22:00, 16 hours) and nighttime (22:00–6:00, 8 hours) noise levels, which we combine to construct a 24-hour equivalent noise level as follows:

$$L_{24h} = 10 \log_{10} \left[\frac{16}{24} 10^{L_{\text{day}}/10} + \frac{8}{24} 10^{L_{\text{night}}/10} \right], \tag{1}$$

where L_{24h} denotes the 24-hour equivalent noise level, and L_{day} and L_{night} denote daytime and nighttime equivalent noise levels, respectively. Following the MOE's environmental standards for areas adjacent to arterial roads, we also construct separate indicators for daytime noise above 70 dB and nighttime noise above 65 dB (MOE, 1998).

3.2 Road Environmental Census

We also use the Road Environmental Census, which collects information on roadside noise levels and noise-control infrastructure. The Ministry of Land, Infrastructure, Transport and Tourism (MLIT) designed the Census to help road authorities decide where and how to reduce road-traffic noise (MLIT, 2004). Through a freedom-of-information request to MLIT, we obtain data covering national roads from 2000 to 2010.[17] For each road segment, the Census records pavement type: dense-graded asphalt, noise-proof asphalt (porous), other asphalt, or concrete. It also records the installation of roadside buffer strips, noise barriers, sound-absorbing panels on elevated structures, and low-noise porous pavement.[18] MLIT measures noise levels using methods similar to those of the CMRTN. Appendix Section A-2 describes how we construct the panel.

[17] The request was made under Japan's Act on Access to Information Held by Administrative Organs. Like the U.S. Freedom of Information Act, this law allows the public to request government records.

[18] Roadside buffer strips separate traffic lanes from nearby homes, allowing noise to weaken with distance. Noise barriers are walls that block the direct path of traffic noise. Sound-absorbing panels beneath elevated roads reduce noise reflected from the underside of the structure. Low-noise porous pavement is asphalt with small air gaps that help reduce noise from contact between tires and the road.

3.3 Land Prices

We obtain residential land values from Japan's Public Notice of Land Prices, which MLIT publishes annually (MLIT, 2010). The data report appraised land values per square meter and exclude the value of buildings. For each site, at least two licensed real-estate appraisers independently estimate its value using recent transactions, with prices adjusted to January 1. The Land Appraisal Committee reviews these appraisals and determines the site's official value under ordinary market conditions. We focus on residential sites to study households' willingness to pay for quieter environments. We construct a balanced panel of residential sites from 2000 to 2010.

3.4 Other Data

We use additional data for each analysis, as described in Appendix A-3. In Section 4, we study noise trends using data on local socioeconomic and demographic conditions (SBJ, 2000). In Section 5, we measure traffic conditions using the Road Traffic Census and annual mileage data from MLIT (MLIT, 2021, 2009, 2022). We use AIRIA vehicle counts to calculate the share of hybrid and plug-in hybrid vehicles (AIRIA, 2025a,b), and Zenkeijikyo sales and vehicle counts to estimate how quickly owners replace older vehicles (Zenkeijikyo, 2025). In Section 6, we use air pollution data from the National Institute for Environmental Studies to examine whether pavement improvements also affect air quality (NIES, 2025).

3.5 Analytical Samples and Summary Statistics

Our analysis uses two main datasets. To examine changes in noise exposure and who benefited from them, we match CMRTN noise measurements from 2002–2022 with local demographics. The CMRTN noise data contain 46,544 observations. Panel A of Appendix Table A1 reports an average 24-hour noise level of 66.4 dB. The standard deviation of 24-hour noise is 4.5 dB. Daytime noise is 4.9 dB louder on average than nighttime noise, averaging 67.4 dB versus 62.5 dB. Approximately 33% of daytime observations and 39% of nighttime observations exceed their respective environmental standards. The average local college-graduate share is 21.3%, while the average population shares of children aged 0–14 and adults aged 65 or older are 13.9% and 17.5%, respectively.

We use the Road Environmental Census from 2000–2010, matched with residential land prices, to estimate residents' willingness to pay for quieter roads. Panel B of Appendix Table A1 reports an average land price of ¥135,000 per square meter and an average 24-hour noise level of 68.2 dB. Low-noise pavement is the most prevalent noise-control measure, with average coverage of 41.7%, followed by noise barriers at 10.8%. Roadside buffer strips and sound-absorbing panels are less common, with average coverage of 3.0% and 0.9%,

respectively. We also match the Census with traffic volume, travel speed, and heavy-vehicle shares from the Road Traffic Census to control for traffic conditions.

4 Trend

4.1 Econometric Model

We estimate changes in road-traffic noise within road segments using the following regression:

$$Y_{it} = \sum_{\tau=2003}^{2022} \delta_\tau \mathbf{1}\{t = \tau\} + X_{it}'\gamma + \mu_i + \varepsilon_{it}, \tag{2}$$

where Y_{it} denotes road-traffic noise for road segment i in year t. We use 24-hour equivalent noise as the main outcome. We also examine daytime and nighttime noise levels, as well as indicators for whether daytime or nighttime noise exceeds the relevant environmental standard. Road-segment fixed effects μ_i allow us to compare noise outcomes within the same road segment over time. These fixed effects absorb time-invariant determinants of noise pollution specific to road segment i, including cross-sectional differences in baseline noise pollution. The year coefficients δ_τ capture average within-road-segment changes in the outcome. We omit the 2002 year indicator, so the coefficients for 2003–2022 measure changes in the outcome relative to 2002. The vector X_{it} captures measurement conditions, including the distance between the monitoring point and the road, the ground height of the monitoring point, and the month of measurement. Standard errors are clustered at the road-segment level.

We summarize the rate of change using parametric time trends:

$$Y_{it} = \alpha f(t - 2002) + X_{it}'\gamma + \mu_i + \varepsilon_{it}, \tag{3}$$

where $f(t - 2002)$ is a linear or quadratic function of year, relative to 2002. The linear specification estimates the average annual change in the outcome over the sample period. The quadratic specification allows the rate of change to vary over time. It captures acceleration or deceleration in the outcome trajectory.

4.2 Noise Levels

We first document the spatial distribution of road-traffic noise and find substantial variation across municipalities. Appendix Figure B1 maps municipality-level averages of noise levels and environmental-standard exceedance rates over 2002–2022, using measurements from

monitored road segments. In particular, two patterns emerge. First, noise levels are relatively high in Japan's three largest metropolitan areas—Tokyo, Osaka, and Nagoya—with lower levels in some municipalities elsewhere in the country. However, high noise levels also occur outside these metropolitan areas. Second, daytime and nighttime noise follow similar spatial patterns (Panels B and C), although daytime levels are higher overall. The shares of observations exceeding the daytime and nighttime environmental standards also show similar geographic patterns (Panels D and E).

We next examine baseline differences in noise exposure across neighborhoods using cross-sectional regressions (Table 1). We use the local college-graduate share as a proxy for socioeconomic status because Japan does not disclose geographically detailed income data. We also examine the local shares of children and older adults because these groups are more vulnerable to environmental pollution.

We do not find a statistically significant association between educational attainment and baseline 24-hour noise exposure. In Table 1, Column (1), the coefficient on the local college-graduate share is positive at 0.63, suggesting higher noise levels in neighborhoods with more college graduates, but the estimate is small and statistically insignificant. The coefficient becomes negative as additional controls enter. In Column (5), which includes measurement conditions, age composition, population density, an urban-area indicator, and prefecture fixed effects, the coefficient is -1.70 and remains statistically insignificant.[19] Comparing neighborhoods at the 75th and 25th percentiles of the local college-graduate share—26.4% and 17.2%, respectively—implies 0.16 dB lower noise in the neighborhood with higher educational attainment. This corresponds to only about 1.1% lower perceived loudness.

We also find no evidence that neighborhoods with larger local shares of children or older adults experienced higher baseline noise exposure. In Table 1, Column (2), higher local shares of residents aged 0–14 and 65 and over are associated with lower noise levels, relative to the omitted working-age population. These associations persist after adding educational attainment in Column (3), population density and the urban-area indicator in Column (4), and prefecture fixed effects in Column (5).

4.3 Noise Trends

The raw data show a substantial decline in road-traffic noise between 2002 and 2022. The average 24-hour equivalent noise level fell by 3.6 dB, from 68.8 to 65.2 dB, with declines in both daytime and nighttime noise (Appendix Figure B2, Panels A–C). The shares of

[19]The urban-area indicator equals one for areas in the Tokyo metropolitan area (Tokyo, Saitama, Chiba, and Kanagawa), the Osaka metropolitan area (Osaka, Kyoto, Hyogo, and Shiga), or the Nagoya metropolitan area (Aichi, Gifu, and Mie).

observations exceeding the corresponding environmental standards also fell markedly (Panels D and E). We compare noise distributions from the first five years of our data (2002–2006) with those from the last five years (2018–2022). Both the upper and lower portions of the distributions shifted toward lower noise levels (Appendix Figure B3). Because the set of monitored road segments changes over time, these patterns may reflect both changes in sample composition and changes in noise levels within segments.

We estimate trends in road-traffic noise by comparing noise levels within the same road segments over time and find a substantial decline. We estimate equation (2), which includes road-segment fixed effects and controls for measurement conditions. Panel A in Figure 3 shows that the 24-hour equivalent noise level fell by 2.84 dB between 2002 and 2022. We find larger declines in noise levels in 2020 and 2021, which may reflect reduced road traffic during the COVID-19 pandemic. This 2.84 dB decline corresponds to about a 48% reduction in acoustic energy and an 18% reduction in perceived loudness. To put this magnitude in context, FHWA (1995) describes a 3 dB reduction as "barely perceptible"—a small change listeners may just notice. It is roughly comparable to halving traffic from 1,000 to 500 vehicles per hour or lowering the speed limit from 50 to 30 km/h (FOEN, 2025). Similar 3 dB differences appear between a robot vacuum's standard and quiet modes (Hales, 2024), between station ticket gates and coffee shops (69 versus 66 dB) (Sueoka et al., 2009), and between the tested 2019 and 2024 Tesla Model 3 interiors at 70 mph (70 versus 67 dB) (VanderWerp, 2024).

Road-traffic noise declined more rapidly early in our study period and more slowly later on. Table 2, Panel A, reports the coefficient on the linear year trend in equation (3). On average, 24-hour equivalent noise fell by 0.13 dB per year, corresponding to roughly a 3% annual reduction in acoustic energy. When we include a quadratic trend (Panel B), the coefficient on year is negative and that on year squared is positive, indicating that the decline slowed over time. The predicted annual decline in noise falls from 0.22 dB in 2002 to 0.13 dB in 2012 and 0.04 dB in 2022. The quadratic trend in Figure 3, Panel B, implies reductions of 1.77 dB between 2002 and 2012 and 0.85 dB between 2012 and 2022, with roughly two-thirds of the predicted decline occurring in the first half of the study period.

Road-traffic noise declined during both daytime and nighttime, with larger reductions at night. By 2022, daytime noise had fallen by 2.63 dB and nighttime noise by 3.91 dB relative to 2002 (Appendix Figure B4, Panels A and B). The larger decline in nighttime noise is particularly relevant for health because nighttime noise can disrupt sleep (WHO, 2018). The probability of exceeding the corresponding environmental standards also fell substantially, by 30.3 percentage points during the daytime and 32.0 percentage points at night (Appendix Figure B4, Panels C and D). These declines are large relative to the initial exceedance rates: in 2002–2006, 51.8% of daytime observations exceeded 70 dB and 59.4%

of nighttime observations exceeded 65 dB. Both noise levels and exceedance probabilities follow a similar pattern, declining more rapidly early in the study period and more slowly later on (Table 2, Panel B).

The decline in road-traffic noise was broadly similar across neighborhoods with different demographic and socioeconomic characteristics.[20] We split road segments at the median of each neighborhood characteristic measured in 2002–2006 and estimate year trends separately for the two groups (Figure 4). The estimated declines are similar in neighborhoods with higher and lower shares of college graduates, children, and older residents. Urban and rural areas, as well as more and less densely populated areas, also experienced similar declines.

In contrast, road segments with initially higher noise levels experienced larger noise reductions. In road segments with initially high noise levels, noise declined by an average of 0.19 dB per year, while segments with low initial noise declined by about 0.07 dB per year (Figure 4). For illustration, noise drops from roughly 71 to 67 dB over 20 years in initially noisier locations, compared with 65 to 64 dB in initially quieter locations. These patterns suggest convergence in road-traffic noise across locations.

These results are largely robust to alternative specifications and sample restrictions. In Appendix Table B1, we use acoustic-energy and standardized z-scores as alternative outcome measures. We also weight regressions by evaluation-section length so that the estimates reflect noise reductions at a representative location along monitored roads rather than on a representative road segment. Alternative samples include a sample of road segments observed at least four times, a sample of segments observed early in the period, and specifications excluding the COVID-19 years 2020–2021. Across these checks, the estimated annual decline in 24-hour equivalent noise remains close to the baseline estimate of 0.13 dB per year, and the corresponding declines in daytime noise, nighttime noise, and environmental-standard exceedance rates are similarly stable. Quadratic trend specifications also deliver the same qualitative conclusion: noise declined more rapidly earlier in the sample and more gradually later on.

5 Causes of Noise Reduction

We have established that road-traffic noise declined substantially between 2002 and 2022. This raises a natural question: what contributed to this decline? In particular, to what extent did Japan's national noise-control policies—tighter vehicle noise-emission standards

[20] Studies of the Clean Air Act find larger improvements in air quality in lower-income and Black communities (Bento et al., 2015; Currie et al., 2023). In their study of the Safe Drinking Water Act, Keiser et al. (2026) document faster declines in drinking-water pollution in low-income communities, but less clear differences by race and ethnicity.

and expanded noise-mitigation infrastructure—contribute? Because the government implemented many of these policies nationwide and some data are available only at the national or prefectural level, it is difficult to identify each policy's effect separately. We therefore use an engineering model to quantify these policies' contributions to noise reduction. The model combines engineering estimates of each intervention's effect with changes in its prevalence.

5.1 Model Inputs and Stylized Facts

Our engineering prediction model accounts for four factors that can affect road-traffic noise: traffic conditions, conventional hybrid and plug-in hybrid adoption, vehicle noise-emission standards, and noise-mitigation infrastructure. Appendix Table C1 summarizes the model's national inputs. Because not all data are available for every year, we interpolate between observed years or extrapolate beyond them where necessary. Appendix C-1 describes these procedures in detail.

First, changes in Japan's traffic conditions may have contributed to the decline in road-traffic noise (Panel A of Appendix Table C1). Between 2002 and 2022, national daily vehicle-kilometers fell by 6.9%, and the heavy-vehicle share fell from 14% to 12%, while average speed changed little overall (Appendix Figure C1). Because each vehicle generates sound and heavy vehicles are generally louder than light vehicles, lower traffic volume and a smaller heavy-vehicle share can reduce road-traffic noise.

Second, the combined share of conventional hybrids and plug-in hybrids (HV/PHV) in Japan's passenger-vehicle stock rose from 0.14% in 2002 to 17% in 2022 (Panel B of Appendix Table C1), following Toyota's launch of the Prius in 1997 (Toyota, 1997).[21] Because conventional and plug-in hybrids can operate on electric power with their combustion engines off, their adoption can reduce road-traffic noise, particularly at low speeds (NHTSA, 2016). We consider conventional and plug-in hybrids only among light vehicles because, in March 2022, battery-electric vehicles represented just 0.2% of Japan's vehicle stock, while conventional hybrids accounted for only 1% of trucks (excluding kei vehicles and trailers) and 0.7% of buses (CEV, nd; AIRIA, 2022; SBJ, 2026).

Third, tighter vehicle noise-emission standards can reduce road-traffic noise. Japan tightened noise-emission standards for passenger cars in 1998–1999 and heavy trucks in 2001. Because these standards apply to new vehicles, their effects emerge gradually as owners replace older vehicles. We estimate that the share of vehicles subject to the tighter standards increased from 36% in 2002 to 92% in 2022 for light vehicles and from 8.8% to 88% for heavy vehicles (Panel C of Appendix Table C1). By 2022, the tighter standards covered roughly

[21] Conventional hybrids account for about 98% of the combined HV/PHV stock reported by AIRIA in March 2022 (AIRIA, 2022).

nine in ten vehicles in both groups.

Lastly, noise-mitigation infrastructure expanded (Panel D of Appendix Table C1). Because road-infrastructure data are available only for 2000–2010, we project each prefecture's infrastructure coverage through 2022 before averaging across prefectures using noise-monitoring counts. Low-noise pavement was the most widespread measure: its national coverage rose from 28% in 2002 to a projected 68% in 2022. Between these years, noise-barrier coverage rose from 8.3% to 13%, buffer-strip coverage from 2.3% to 3.7%, and sound-absorbing panel coverage from 0.22% to 1.2% (Appendix Figure C2).

5.2 Method

5.2.1 Noise Prediction Model

To predict road-traffic noise, we build a simple model based on the sound-energy framework of the ASJ RTN-Model (Sakamoto et al., 2019; Okada et al., 2019), which MOE guidance recommends for assessing noise levels and designing noise-mitigation policies.[22] The ASJ model predicts roadside noise by combining sound energy from different vehicle types and accounting for traffic volume, speed, and road infrastructure. We extend this framework to incorporate tighter vehicle noise-emission standards and changes in the conventional hybrid and plug-in hybrid (HV/PHV) share.

We first predict the acoustic energy generated by road traffic as follows:

$$E_t^{\text{total}} = Q_t^{\alpha_Q} v_t^{\gamma} \left[\underbrace{(1-\omega_t) G_t^S H_t}_{\text{Light vehicles}} + \underbrace{\omega_t 10^{\delta_L/10} G_t^L}_{\text{Heavy vehicles}} \right] \underbrace{10^{-\frac{1}{10}\sum_{k=1}^{4} d_k c_t^k}}_{\text{Road infrastructure}}. \tag{4}$$

Here, E_t^{total} is the predicted national sound-energy index in year t. The variable Q_t denotes an index of national daily vehicle-kilometers, normalized to 100 in 2002, and v_t denotes average traffic speed. The parameters α_Q and γ are the sound-energy elasticities with respect to traffic volume and speed. The heavy-vehicle share is ω_t, and the light-vehicle share is $1-\omega_t$. The parameter δ_L is the additional noise generated by a heavy vehicle relative to a light vehicle, in decibels; $10^{\delta_L/10}$ converts this difference into a sound-energy ratio.

We use G_t^S and G_t^L to capture how tighter standards reduce the sound energy emitted by light and heavy vehicles, respectively. We define them as

$$G_t^S = (1 - s_t^S) + s_t^S 10^{-\theta_R d_s/10}, \qquad G_t^L = (1 - s_t^L) + s_t^L 10^{-\theta_R d_L/10}.$$

[22] Europe's CNOSSOS-EU and the U.S. Federal Highway Administration's Traffic Noise Model use the same basic approach (Kephalopoulos et al., 2012; Menge et al., 1998).

The noise reduction from tighter emission standards depends on three elements. First, s_t^S and s_t^L measure the shares of light and heavy vehicles subject to the tighter standards. As owners replace older vehicles with new ones, more vehicles become subject to these standards. Second, d_s and d_L measure how much the standards tighten noise limits for each vehicle group, in decibels. Third, some vehicles already meet the tighter limits, and noise reductions under test conditions may differ from those during everyday driving. We therefore introduce θ_R to capture how much tighter limits reduce vehicle noise in practice.

The factor H_t captures how replacing conventional light vehicles with quieter hybrids and plug-in hybrids reduces average sound energy:

$$H_t = (1 - h_t) + h_t 10^{-d_H/10}.$$

We set the sound energy emitted by a conventional vehicle equal to one. A hybrid or plug-in hybrid emits $10^{-d_H/10}$ times this amount, where d_H is its predicted noise reduction in decibels. Here, h_t is the hybrid and plug-in hybrid share of passenger vehicles.

We account for four types of noise-mitigation infrastructure: low-noise pavement, noise barriers, buffer zones, and sound-absorbing panels. Each type k reduces predicted noise by $d_k c_t^k$, where d_k is the reduction in decibels on covered roads and c_t^k is national coverage formed by weighting prefectural covered-length shares by noise-monitoring counts.

We calculate the predicted change in noise relative to 2002 by converting the change in sound energy into decibels:

$$\Delta \widehat{L}_t = 10 \log_{10} \left(\frac{E_t^{\text{total}}}{E_{2002}^{\text{total}}} \right).$$

To calculate each factor's contribution to the predicted noise change, we change the inputs one at a time.[23] We then average each factor's contribution across all possible orders of these changes.

The predicted noise change is the sum of contributions from traffic conditions, hybrid adoption, tighter vehicle standards, and road infrastructure.

$$\Delta \widehat{L}_t = \underbrace{\Delta L_t^{\text{traffic}}}_{\text{Traffic conditions}} + \underbrace{\Delta L_t^{\text{HV/PHV}}}_{\text{Hybrid adoption}} + \underbrace{\Delta L_t^{\text{regulation}}}_{\text{Vehicle standards}} + \underbrace{\Delta L_t^{\text{infra}}}_{\text{Road infrastructure}}. \tag{5}$$

[23] We use a Shapley decomposition to separate the interacting contributions of traffic conditions, vehicle standards, and hybrid adoption, averaging each factor's contribution across all possible orders of changing inputs from 2002 to year t. Infrastructure contributions equal coverage changes multiplied by the corresponding noise-reduction parameters. Together, the contributions sum to the predicted noise change.

5.2.2 Model Parameters

Table 3 summarizes the model parameters. For traffic conditions (Panel A), following the ASJ framework, we set the sound-energy elasticities with respect to traffic volume and speed to $\alpha_Q = 1$ and $\gamma = 1$, respectively (Sakamoto et al., 2019; Okada et al., 2019). This means that a 1% increase in either traffic volume or speed raises predicted sound energy by 1%, holding other inputs fixed. We set $\delta_L = 7.4$ dB, meaning that a heavy vehicle generates 7.4 dB more noise than a light vehicle, or about 5.5 times as much acoustic energy (Okada et al., 2019).

Weighting the changes in noise limits by vehicle categories' production shares, we use a 2 dB tightening for light vehicles ($d_s = 2.0$) and a 2.6 dB tightening for heavy vehicles ($d_L = 2.6$; Panel C) (MOE, nda,n). We also use $\theta_R = 0.5$, implying that a 1 dB tightening of noise limits reduces vehicle noise on the road by about 0.5 dB, following Sandberg's (2001) evidence.

We assume that HVs and PHVs emit 1.5 dB less noise than comparable conventional vehicles ($d_H = 1.5$ dB; Panel B), equivalent to about 29% less acoustic energy (Okada et al., 2020; Yamauchi, 2025).

For road infrastructure (Panel D), we use a noise reduction of 2.5 dB for low-noise pavement, consistent with our analysis in Section 6.2. We use reductions of 5 dB for noise barriers, 6 dB for buffer zones, and 3 dB for sound-absorbing panels, following the expected noise reductions in MLIT and FHWA guidance (MLIT, 2003; FHWA, 2000).

5.3 Results

5.3.1 Main Results

The engineering predictions align well with the regression estimates (Figure 5). Our engineering model predicts a 2.91 dB decline in road-traffic noise between 2002 and 2022, compared with a 2.84 dB decline in the regression analysis (Table 4, column (5)). On average across years, the model predicts a noise decline 0.02 dB smaller than that implied by the regression year effects. The average absolute gap is 0.16 dB (Appendix Table C2). Additionally, the model predicts that noise declines by 1.72 dB during 2002–2012 and by 1.20 dB during 2013–2022 (Table 4, columns (1) and (3)). About 59% of the decline occurs in the first half of the study period, consistent with the larger early decline in our regression results.

Traffic conditions account for 22% of the predicted noise decline. HV/PHV adoption contributes 4.2% (Table 4, Panels A and B). Among changes in traffic conditions, lower traffic volume contributes most to the predicted noise decline, accounting for 0.31 dB, or 11% of the total decline (Panel A). Additionally, lower traffic volume produces larger predicted noise reductions in 2020 and 2021, when the COVID-19 pandemic reduced road traffic. A

smaller heavy-vehicle share further reduces predicted noise by 0.27 dB, or 9.1% of the total decline, and lower speed by 0.07 dB. In contrast, HV/PHV adoption contributes only 0.12 dB, or 4.2%, of the predicted decline (Panel B). Most of this contribution emerges after 2010 as adoption increases.

Tighter vehicle noise-emission standards contribute 0.79 dB, or 27%, of the predicted decline in road-traffic noise (Table 4, Panel C). Most of the noise reduction from tighter standards occurs in the first half of the study period. By 2012, they reduce predicted noise by 0.58 dB, or about 74% of their total contribution. By then, 81% of light vehicles and 66% of heavy vehicles are already subject to the tighter standards. In contrast, further gains after 2012 are limited because most vehicles are already subject to the tighter standards, leaving fewer older vehicles to replace.

Noise-mitigation infrastructure contributes substantially to the predicted noise decline. Table 4, Panel D, shows a contribution of 1.35 dB, or 46% of the total decline. Low-noise pavement is the most widely adopted noise-mitigation intervention and accounts for most of this reduction, contributing 1.00 dB, or 34% of the total predicted decline. Most of the predicted noise reduction from low-noise pavement occurs in the first half of the study period, reflecting its rapid expansion from 28% coverage in 2002 to 57% in 2010. Noise barriers contribute another 0.24 dB, or 8.1%. Sound-absorbing panels and buffer strips make smaller contributions of 0.03 and 0.08 dB, respectively, as their coverage is limited to a few percent of roads.

Taken together, these results suggest that Japan's noise-mitigation policies explain much of the observed decline in road-traffic noise between 2002 and 2022. In the engineering model, tighter vehicle noise-emission standards and expanded noise-mitigation infrastructure account for 2.14 dB, or 73%, of the predicted decline (Table 4, column (5)). Among these policies, low-noise pavement alone accounts for about one-third of the decline.

Japan's noise-mitigation policies contributed most to the predicted noise decline during the first half of the study period. By 2012, these policies accounted for 1.69 dB, or 98%, of the 1.72 dB decline in the engineering model (Table 4, column (1)). During 2013–2022, they accounted for 0.45 dB, or 38%, of the decline, while changes in traffic conditions accounted for 0.65 dB, or 54% (Table 4, column (3)). These policies contributed more in the first half because noise-mitigation infrastructure expanded rapidly and vehicle owners replaced older vehicles with newer ones subject to tighter emission standards. After 2012, infrastructure expansion slowed, and tighter standards already covered most vehicles, leaving less scope for further noise reductions. Instead, lower traffic volumes and speeds accounted for much of the predicted noise decline during 2013–2022.

5.3.2 Sensitivity Analysis

We assess how alternative parameter assumptions change the decomposition results, using the values in Appendix Table C3. The results remain qualitatively similar across these assumptions (Appendix Table C4).

For example, suppose we use the lower values for the vehicle-standard parameters, holding all other parameters at their baseline values. We reduce the assumed tightening of noise limits from 2.0 to 1.8 dB for light vehicles (d_s) and from 2.6 to 2.2 dB for heavy vehicles (d_L). We also assume that each 1 dB tightening of the noise limit reduces vehicle noise on the road by 0.4 dB ($\theta_R = 0.4$), compared with 0.5 dB in the baseline model. These assumptions reduce the predicted contribution of vehicle standards to 0.55 dB, compared with 0.79 dB at baseline. Similarly, lowering the assumed noise reduction from low-noise pavement (d_P) from 2.5 to 2.0 dB, while holding all other parameters at baseline, reduces its contribution to 0.80 dB, compared with 1.00 dB at baseline.

When we use the lower values for both interventions simultaneously, holding all other parameters at baseline, they jointly reduce noise by 1.35 dB out of a total predicted decline of 2.47 dB, or 55%. These policies still account for more than half of the predicted decline. Using their upper values instead raises their combined contribution to 2.35 dB out of a total predicted decline of 3.48 dB, or 68%.

6 Noise, Infrastructure, and Land Prices

We have shown that noise pollution declined substantially over the two decades and that Japan's national policies account for the majority of this decline. How much do residents near roads gain from such policy-induced noise reduction? The answer depends on how much they value quieter roads. We estimate how conversion from dense to low-noise pavement affects noise levels and residential land prices. Dividing the land-price effect by the noise reduction gives a two-sample instrumental variables estimate of households' willingness to pay per decibel, expressed as a percentage of land prices.

6.1 Econometric Model

We estimate the impact of resurfacing from dense-graded to low-noise pavement on noise with

$$Y_{st} \;=\; \beta\, D_{st} \;+\; X_{st} \;+\; \mu_s \;+\; \delta_t \;+\; \varepsilon_{st}, \tag{6}$$

where Y_{st} is the 24-hour equivalent noise level for road segment s in year t, and D_{st} equals one if s has been resurfaced from dense-graded to low-noise pavement by year t. Hence, the

treatment group includes segments that converted from dense-graded to low-noise pavement between 2000 and 2010, and the control group includes segments that did not convert to low-noise pavement in this period, including segments that were already low-noise when first observed. [24] Identification comes from the change in noise at resurfaced segments relative to segments never resurfaced to low-noise pavement. X_{st} represents controls for the road width, the number of lanes, and distance to the road center. Segment fixed effects μ_s absorb time-invariant segment characteristics, and year fixed effects δ_t absorb shocks common to all segments. β is our parameter of interest and measures the effect of pavement conversion on noise levels, in decibels. We cluster standard errors by segment.

We assess the pre-trend and dynamic pattern of the treatment effect with the event-study analogue of (6),

$$Y_{st} \;=\; \sum_{\substack{k=-5 \\ k\neq -1}}^{5} \beta_k \, \mathbf{1}\{t - g_s = k\} \;+\; X_{st} \;+\; \mu_s \;+\; \delta_t \;+\; \varepsilon_{st}, \tag{7}$$

where g_s is the year when resurfacing from dense-graded to low-noise pavement takes place in segment s, k is event time, and segments never resurfaced from dense to low-noise pavement have all event-time indicators equal to zero. $k = -1$ is the omitted period. Coefficients close to zero for $k < 0$ suggest similar pre-treatment trends between treated and comparison segments. Coefficients for $k \geq 0$ trace how the effect of pavement conversion on noise evolves over time.

We use the heterogeneity-robust estimator of Callaway and Sant'Anna (2021) in our main analysis because pavement conversion occurred in different years and its effects may vary across road segments and over time. Conventional two-way fixed effects can assign negative weights to some treatment effects under staggered adoption, potentially biasing the estimated average effect when treatment effects vary across road segments or over time (Goodman-Bacon, 2021; De Chaisemartin and d'Haultfoeuille, 2020).

We estimate the effect of pavement conversion on land prices using a difference-in-difference-in-differences (DDD) design that uses distance to the road as a third difference. We exploit the difference in distance to roads because traffic noise fades with distance. Reported noise annoyance declines sharply over the first 150m from major roads, and traffic noise can become nearly inaudible at around 500m from the road (Ragettli et al., 2016; Wilk et al., 2025). We therefore use land parcels in the 500–1000m band as the comparison group. We compare the effect of pavement conversion on land prices in the 0–100m, 100–200m, and 200–500m bands with its effect on land parcels 500–1000m from the same road segment.

[24] We exclude segments that converted to low-noise pavement from other asphalt or concrete.

Intuitively, this comparison accounts for broader amenity improvements shared by nearby and more distant properties. If these improvements affect both groups similarly, the difference in land price responses captures the more localized benefits of noise reduction. Our identifying assumption is that, without pavement conversion, the land-price gap between near and comparison parcels would have evolved similarly along treated and untreated road segments.

We estimate:

$$\ln P_{it} = \rho D_{it} + \tau D_{it}\mathbf{1}\{i \in \mathcal{I}_b\} + X_{it} + \lambda_i + \delta_t + \theta_t \mathbf{1}\{i \in \mathcal{I}_b\} + \varepsilon_{it}, \tag{8}$$

where P_{it} is the appraised price per square meter of parcel i in year t, and D_{it} equals one if the road segment nearest parcel i has been converted to low-noise pavement by year t. The indicator $\mathbf{1}\{i \in \mathcal{I}_b\}$ equals one for parcels in distance band b and zero for parcels in the 500–1000m comparison band. We estimate the model separately for the 0–100m, 100–200m, and 200–500m bands. X_{it} represents the road width and the number of lanes the road segment nearest parcel i in the earliest year observed. Parcel fixed effects λ_i absorb time-invariant parcel characteristics. Year fixed effects δ_t and their interactions with the distance-band indicator, $\theta_t \mathbf{1}\{i \in \mathcal{I}_b\}$, allow the two bands to follow different time trends. The parameter of interest τ is the price effect in band b relative to the 500–1000m comparison band.

We implement a corresponding event study with the following specification:

$$\ln P_{it} = \sum_{\substack{k=-5 \\ k\neq -1}}^{5} \big(\rho_k + \tau_k \mathbf{1}\{i \in \mathcal{I}_b\}\big)\mathbf{1}\{t - g_i = k\} + X_{it} + \lambda_i + \delta_t + \theta_t \mathbf{1}\{i \in \mathcal{I}_b\} + \varepsilon_{it}. \tag{9}$$

Here, g_i is the year the road segment nearest parcel i was converted from dense to low-noise pavement. Parcels matched to the same road segment share the same g_i. We mainly report a triple difference estimator proposed by Ortiz-Villavicencio and Sant'Anna (2025), which is a triple difference analog of Callaway and Sant'Anna (2021), alongside two-way fixed effects results. We cluster standard errors by road segment.

6.2 Effect of Pavement Conversion on Noise Levels

Figure 6 presents event-study estimates based on equation (7). The horizontal axis displays relative years from pavement conversion, with the year before conversion set as the reference year. We report two-way fixed effects coefficients with gray triangles and Callaway-Sant'Anna coefficients with black circles.

Pavement conversion from dense to low-noise pavement reduces noise levels immediately, but the effect diminishes over time. Figure 6 shows that pre-treatment coefficients are

individually statistically indistinguishable from zero at the 5% level. [25] In the year of treatment, noise levels immediately drop by 3.49 dB, and the effect becomes 1.95 dB by year 5 (Callaway–Sant'Anna estimator). Thus, the effect seems to attenuate by 0.31 dB per year on average.

Using the Callaway–Sant'Anna estimator, we estimate an average noise reduction of 2.55 dB over five years (Table 5, Panel A, Column (1)). The two-way fixed-effects estimate implies a reduction of 3.14 dB (Panel C, Column (1)). A 2.55 dB reduction corresponds to a 44 percent decrease in acoustic energy, or a 16 percent decrease in perceived noise.

Our estimated 3.49 dB noise reduction in the year of conversion is close to existing engineering benchmarks. Mikhailenko et al. (2022) report typical initial reductions of 3–3.5 dB for porous asphalt compared with dense asphalt concrete, while Japan's Environment Agency (2000) reports about 3 dB. Our smaller five-year average reduction of 2.55 dB is consistent with evidence that the effect weakens as pavement ages (Bendtsen et al., 2010; Xu et al., 2024). Over time, dirt and debris clog the pavement's pores. This reduces sound absorption and prevents air from escaping as easily from beneath passing tires, increasing tire-road noise.

6.2.1 Effects on Air Pollution

We test whether pavement conversion affects air pollution. Road traffic generates both noise and air pollution, and policies that reduce one may also affect the other. If pavement conversion somehow changes air quality, although this seems unlikely, these changes could contribute to the estimated land-price effects. We therefore examine whether conversion affects nitrogen oxides (NOx) and carbon monoxide (CO), two major pollutants emitted by vehicles, as well as suspended particulate matter (SPM).

Pavement conversion has small, statistically insignificant effects on air pollution, with mixed signs. Table D1 shows small estimated changes relative to pre-treatment means: an increase of 1.26 ppb (1.9%) in NOx, a decrease of 1.20 $\mu g/m^3$ (3.4%) in SPM, and an increase of 25.2 ppb (3.4%) in CO. The signs are mixed, and none of these estimates is statistically significant at the 10 percent level.

Note that air-pollution monitoring stations are sparsely distributed in Japan. This limits our ability to directly control for local air pollution when estimating the effects of pavement conversion on noise levels and land prices.

[25] The pre-treatment coefficients are also jointly insignificant: a Wald test that the coefficients for years -5 to -2 are all zero gives $p = 0.26$ for the Callaway–Sant'Anna estimates and $p = 0.72$ for two-way fixed effects.

6.3 Effect of Pavement Conversion on Land Prices

We report DDD event studies for the 0–100m, 100–200m, and 200–500m bands relative to the 500–1000m comparison band, based on equation (9). We report both two-way fixed-effects and Ortiz-Villavicencio–Sant'Anna estimates, treating the latter as our baseline results.

Figure 7 shows that converting dense pavement to low-noise pavement increases land prices for parcels near the road relative to those farther away. The effects are concentrated among parcels within 100m of the road (Panel A). Before conversion, the land-price gap between nearby and more distant parcels follows similar trends around treated and untreated roads. After conversion, the estimated relative price gain for parcels within 100m grows gradually, reaching approximately 0.06 log points by year 5. The effects emerge with a delay, which may reflect the time residents and buyers need to notice quieter conditions or the slow adjustment of appraised land values.

In contrast, we find no statistically significant effects on land prices in the 100–200m (Panel B) or 200–500m (Panel C) distance bands. This is likely because households farther from the road experience smaller noise reductions. Road traffic noise fades with distance, and buildings provide additional shielding. In Japan, many homes are surrounded by closely spaced buildings, so this shielding can make noise decline more quickly with distance (Fujimoto et al., 2003; Sakamoto et al., 2023). In fact, beyond roughly 100m, our estimated 2.55 dB reduction in roadside noise translates into less than 1 dB, a change FHWA (2021) describes as "generally hard to notice" and therefore unlikely to affect land prices.[26]

Column (2) of Panel B in Table 5 shows that pavement conversion raises land prices within 100m by 3.6% on average during the post-treatment period, relative to properties 500–1000m away. This estimate is statistically significant at the 5 percent level. The estimated effects fall to 0.6% for parcels 100–200m away and 0.0% for parcels 200–500m away (Columns (3) and (4)); neither is statistically significant at the 10 percent level. In Panel C, TWFE estimates that pavement conversion increases land prices within 100m by 3.0%, significant at the 10 percent level. TWFE also yields small, statistically insignificant effects beyond 100m.

6.4 Willingness to Pay for Quieter Roads

We estimate households' willingness to pay for a 1 dB reduction in road traffic noise using two-sample instrumental variables (TSIV). Specifically, we divide the estimated effect of

[26]For illustration, we assume road noise of 70 dB at 10m, background noise of 50 dB, and 10 dB of building shielding. We apply standard line-source attenuation of approximately 3 dB per doubling of distance (FHWA, 2017). Under these assumptions, a 2.55dB reduction in road-generated noise reduces total noise by 1.09dB at 100m, 0.85dB at 150m, 0.70dB at 200m, and 0.33dB at 500m.

pavement conversion on log land prices by the estimated reduction in noise levels, using estimates from the two separate samples. This ratio measures willingness to pay for a 1 dB reduction as a share of land value.

We find that residents within 100m of roads are willing to pay about 1.42 percent of land prices for a one-decibel reduction in road traffic noise. We obtain this estimate by dividing the approximately 3.6 percent increase in land prices (Table 5, Panel B, Column (2)) by the 2.55 dB reduction in noise (Panel A, Column (1)). The standard error is 0.006 log points per decibel.[27] We estimate willingness to pay only for residents within 100m of roads, since the land-price effects beyond this distance are small and statistically insignificant (Figure 7).

This is, to our knowledge, the first estimate of willingness to pay for noise reduction using directly measured noise levels. Existing studies generally estimate how events expected to change noise exposure affect property prices, but do not directly estimate the first-stage effects on measured noise levels (Boes and Nüesch, 2011; Moretti and Wheeler, 2025; Allroggen et al., 2025).

Our estimated willingness to pay of 1.42 percent of land value per decibel of noise reduction is broadly consistent with, though somewhat higher than, previous estimates. For road noise, Moretti and Wheeler (2025) report 0.94 percent per decibel, while von Graevenitz (2018) reports 0.1–1.4 percent. Ahlfeldt et al. (2019) estimate 1.3 percent for rail noise. For aircraft noise, estimates are 0.1 percent (Sugasawa et al., 2025), 0.5 percent (Boes and Nüesch, 2011), 0.6–1.0 percent (Allroggen et al., 2025), and 1.7 percent (Winke, 2017). Our relatively high estimate may partly reflect our use of directly measured noise reductions. If engineering models overstate actual reductions, model-based estimates understate willingness to pay per decibel. Differences in noise sources, exposure measures, and property-market outcomes may also contribute.

6.5 Benefits and Costs of Pavement Conversion

We have shown that conversion to low-noise pavement can increase nearby land prices by reducing noise exposure. Low-noise pavement has been the most widely adopted noise-mitigation measure in Japan since the 2000s. Is this popular approach cost-effective? We compare residents' gains, reflected in higher land prices, with the additional cost of upgrading to low-noise pavement.

We estimate a benefit-cost ratio of 5.0, suggesting that low-noise pavement is highly cost-effective. (See Appendix D-2 for detailed calculations.)[28] We obtain a high benefit-cost

[27] We use the delta method, assuming zero covariance between the estimated effects on noise and land prices.

[28] Even if we count the full cost of low-noise pavement at conversion and every ten years thereafter, the median segment roughly breaks even (benefit-cost ratio 0.97). This is partly because 28 percent of the

ratio, presumably because roads need resurfacing anyway, so reducing noise requires only the additional cost of choosing low-noise pavement. This additional cost is, in fact, modest. Low-noise pavement costs 251 to 358 yen per square meter (about US$0.19 to 0.28 per square foot), a premium of 22 to 32 percent. Low-noise pavement also appears much more cost-effective than noise barriers, Japan's second most widely used infrastructure measure to reduce road traffic noise. Moretti and Wheeler (2025) report a marginal value of public funds of 1.7 for the sound barriers built in their sample, measured as residents' willingness to pay over the net cost to government, which is the same comparison our ratio makes for pavement. Noise barriers appear more expensive because they require new construction, while low-noise pavement adds only a small premium to scheduled resurfacing.

6.6 Pigouvian Tax

A first-best policy for addressing environmental externalities uses taxes to make polluters pay for the harm they impose on others. For road traffic noise, this means setting a tax per kilometer equal to the harm that an extra kilometer of driving causes nearby residents. To obtain this Pigouvian tax, we first calculate the additional noise from one extra kilometer of driving, then monetize the resulting harm using residents' willingness to pay for quieter roads.

We estimate a traffic-weighted average Pigouvian tax of 3.9 yen (US$0.03) per vehicle-kilometer. This corresponds to an average annual tax of 5,700 yen (US$48) per vehicle, or a lifetime tax of 143,000 yen (US$1,200).[29] If we tax light and heavy vehicles separately, the estimated rates are 2.1 yen (US$0.02) and 11.6 yen (US$0.10) per vehicle-kilometer, respectively—a tax more than five times as high for heavy vehicles such as buses and trucks. This large difference reflects the fact that a heavy vehicle generates about 5.5 times as much sound energy as a light vehicle. For comparison, Moretti and Wheeler (2025) use data from Florida in the United States to estimate a capitalized noise cost of US$974 per vehicle, close to our estimate of 143,000 yen (US$1,200). Yet, these similar estimates may arise for different reasons in the United States and Japan. Road traffic noise is more costly in Japan because more people live near roads and residents have a higher willingness to pay for quiet roads. But on average, vehicles in Japan travel much shorter distances, resulting in similar costs per vehicle.

segments do not have any residential land within 100m. Among segments with residential land within 100m, the median ratio is 2.0.

[29] We use a 4% discount rate.

7 Welfare

We showed that residents near major roads are willing to pay for quieter roads. What, then, are two decades of Japan's concerted efforts to quiet its roads worth? 15.5 million households live within 100m of Japan's major roads, with traffic noise at their doorstep. We quantify the benefits of quieter roads for these households using back-of-the-envelope calculations based on gains in residential land values.

Between 2002 and 2022, Japan's noise-control policies raised residential land values near major roads by about 3.0 percent. We calculate this gain by multiplying the 2.14-decibel noise reduction that these policies achieved (Section 5) by residents' willingness to pay 1.42 percent of land value for each one-decibel reduction in noise (Section 6).

To put these gains in perspective, we compare them with the benefits of reducing other pollutants. Japan's noise-control policies generated land-value gains comparable to those that an 8 percent reduction in suspended particulate matter (SPM) concentrations would produce. Existing studies suggest that a 1 percent increase in particulate matter concentrations reduces land prices by about 0.4 percent (Kang et al., 2024; Nishitateno and Burke, 2021).[30] This elasticity of 0.4 implies that reducing SPM concentrations by about 8 percent would produce the same 3.0 percent increase in land prices. In contrast, the gains from quieter roads are smaller than some of the price effects associated with major local environmental hazards. Toxic plant openings reduce housing values within half a mile by about 11 percent (Currie et al., 2015), while Superfund cleanup raises median housing values in nearby census blocks by 14.7 percent (Gamper-Rabindran and Timmins, 2013). Large data centers reduce housing values by 7.2 percent within one mile (Schlauch and Greenhill, 2026).

Over two decades, Japan's efforts to combat road traffic noise pollution generated about 6.8 trillion yen (US$57 billion) in benefits. For the roughly 15.5 million households living within 100m of major roads, these benefits average about 438,000 yen (US$3,700) per household, or 21,900 yen (US$180) per year when spread evenly over 20 years. Spread across Japan's entire population, the benefits amount to about 54,000 yen (US$450) per person. Among them, low-noise pavement accounts for 3.2 trillion yen (US$27 billion) and tighter vehicle emission standards for 2.5 trillion yen (US$21 billion), while other road infrastructure improvements account for the remaining 1.1 trillion yen (US$9 billion).

To our knowledge, this study provides the first nationwide estimates of the welfare ben-

[30] Using the Clean Air Act, Chay and Greenstone (2005) estimate that a 1 percent reduction in particulate pollution raises housing prices by 0.20–0.35 percent. In Japan, Kang et al. (2024) report a 2.74 percent increase in land prices and a 7.17 percent decline in SPM concentrations, implying an elasticity of about -0.38 ($2.74/(-7.17)$). The policy also reduced nitrogen oxides, so this ratio captures the combined air-quality benefit relative to the SPM reduction. Nishitateno and Burke (2021) estimate land-price elasticities with respect to SPM concentrations between -0.4 and -1.0.

efits of road-noise control policies worldwide, and the first such estimates for environmental policies in Japan. In the United States, cost–benefit analysis has long played a central role in evaluating environmental policies. The U.S. Environmental Protection Agency (EPA) estimates that the benefits of the 1990 Clean Air Act Amendments exceed their costs by more than 30 to one (U.S. EPA, 2011). Researchers have also extensively studied the economic consequences of federal laws to clean up rivers and lakes (Keiser and Shapiro, 2019), improve drinking-water safety (Keiser et al., 2026), clean up hazardous waste sites (Greenstone and Gallagher, 2008), and protect biodiversity (Frank et al., 2025), though such efforts remain largely absent for noise control, perhaps because researchers have few policies to assess or limited data to work with.

Japan presents a sharp contrast. Despite more than half a century of environmental regulation—since Japan enacted the Air Pollution Control Act and Noise Regulation Act in 1968, passed the Water Pollution Control Act in 1970, and established the Environment Agency in 1971—the nationwide economic benefits of these efforts remain largely unknown (Botta and Yamasaki, 2020). One possible reason is a reluctance to place a monetary value on the environment. In the 1970s, after pollution had caused severe illness and loss of life, efforts to weigh the costs of environmental protection against human health and life were "subject to strong criticism and condemnation" (Matsuno and Ueta, 2002, pp. 145–146). The government's 1973 Environmental White Paper echoed this view: "A clean environment has great value that *cannot be expressed in monetary terms*." It continued: "Many aspects of it must be protected, *no matter how great the cost to society*" (authors' translation; Environment Agency of Japan, 1973). Since then, this reluctance may have remained part of the government's policy culture for roughly half a century. To date, we are unaware of any study that uses micro-level data and carefully constructed counterfactuals to estimate the nationwide benefits and costs of environmental policies in Japan, including noise-control policies.[31]

Are Japan's noise-mitigation policies—which generated an estimated 6.8 trillion yen (US$57 billion) in benefits over two decades—worth their costs? Several considerations help put these gains into perspective. First, cost-effectiveness depends on the noise intervention. We show that one widely used intervention—low-noise pavement—is worth the cost, with an estimated benefit–cost ratio of 5.0. On the other hand, our empirical analysis does not establish whether vehicle noise-emission standards are worth their costs, even though they contributed substantially to reducing noise pollution. Making vehicles quieter requires

[31] Kochi et al. (2002) estimate the benefits and costs of Japan's air-pollution controls using a modeling approach, without micro-level data or a rigorous empirical design. Kang et al. (2024) and Nishitateno and Burke (2021) use micro-level data to estimate regional benefits in the Tokyo metropolitan area and in Aichi and Mie prefectures, respectively.

automakers to invest in technologies such as improved exhaust systems, engine insulation, and low-noise tires. These potentially costly requirements have, in fact, fueled automakers' resistance to stricter vehicle-noise standards, as debates in Europe illustrate (ACEA, 2012).[32] Evaluating these costs requires estimating welfare effects in the vehicle market beyond automakers' compliance costs, as higher production costs can affect market power, vehicle prices, sales, and both consumer and producer welfare.

It is also useful to compare these benefits with public spending on noise mitigation, although such spending can be difficult to interpret. Japan spent approximately 1.2 trillion yen (US$10 billion) on roadside environmental improvement programs across ten fiscal years between 2000 and 2010 (Board of Audit of Japan, 2004, 2012). A simple comparison suggests substantial net gains from Japan's policies, but this conclusion would be premature. These expenditures often cover environmental improvements beyond noise reduction, such as air-pollution control and roadside tree planting (Ministry of the Environment, Japan, 2001; Ministry of Land, Infrastructure and Transport, Japan, 2007). Additionally, these expenditure figures may not capture other relevant costs. For example, municipalities often install low-noise pavement as part of routine road resurfacing or urban redevelopment, recording the spending under these projects rather than noise mitigation.

Several caveats are important when interpreting our estimates of the benefits of Japan's noise-control policies. First, CMRTN focuses on major roads, including national and prefectural roads and municipal roads with four or more lanes. Residents along roads outside this network may also face substantial noise exposure. If Japan's noise-control policies reduce noise on these unmonitored roads, the total benefits could exceed our estimates.

Second, our willingness-to-pay estimate may not generalize to other roads or interventions. We derive this estimate from low-noise pavement conversions using the Road Environmental Census and apply it to the broader CMRTN network. However, these two datasets cover different road segments, and residents may value noise reductions differently across locations. Moreover, residents' willingness to pay may vary across interventions because they reduce noise exposure in different ways. For example, noise barriers can reduce noise exposure by 5–10 dB, depending on their height (FHWA, 2000).

Third, our calculations may omit benefits beyond those reflected in nearby residential land values. Improvements in engine insulation and tire technology to comply with vehicle noise emission standards may also enhance comfort for drivers and passengers. Quieter road surfaces may generate similar benefits. Moreover, noise-reducing technologies induced by Japanese regulations may benefit residents abroad. For example, quieter vehicles sold in the

[32] ACEA (2012) argued for less stringent limits and longer development lead times, citing investment costs and technical constraints. ACEA (2014) subsequently welcomed the final agreement while emphasizing the technical challenges of meeting the new limits.

United States could reduce noise exposure for American residents (see Ito et al., 2026, for related evidence on international spillovers).

8 Conclusion

This paper provides the first evidence on how national noise-control policies affect noise exposure and residents' welfare. We combine two decades of largely untapped noise measurements from more than 14,000 locations across Japan with detailed policy data obtained through freedom-of-information requests. We document a substantial decline in road-traffic noise: between 2002 and 2022, roadside noise fell by 2.8 decibels, equivalent to roughly a 20% reduction in perceived loudness. Our engineering model attributes approximately three-quarters of the predicted decline to noise-control policies, particularly tighter vehicle noise-emission standards and the expansion of low-noise pavement. We estimate that residents are willing to pay 1.42% of residential land value for a one-decibel reduction in road-traffic noise. Aggregating these benefits, we estimate that Japan's national noise-control policies generated approximately 6.8 trillion yen (US$57 billion) in capitalized benefits over two decades.

References

ACEA (2012). Sound levels of motor vehicles. Industry brochure, European Automobile Manufacturers' Association.

ACEA (2014). ACEA welcomes the conclusion of discussions on the regulation of noise from motor vehicles. European Automobile Manufacturers' Association, April 4.

Ahlfeldt, G. M., Nitsch, V., and Wendland, N. (2019). Ease vs. noise: Long-run changes in the value of transport (dis)amenities. *Journal of Environmental Economics and Management*, 98:102268.

AIRIA (2022). Low-emission vehicle stocks by fuel and vehicle type. Stocks as of March 31, 2022. In Japanese; title translated.

AIRIA (2025a). Historical stocks of hybrid and electric vehicles. March-end stocks; the hybrid table includes plug-in hybrids. Available at https://www.airia.or.jp/publish/file/HVEV_2025.pdf.

AIRIA (2025b). Historical vehicle stocks by vehicle type. March-end stocks, including kei vehicles. Available at https://www.airia.or.jp/publish/file/hoyuudaisuusuii_2025.pdf.

Allroggen, F., Hansman, R. J., Knittel, C. R., Li, J., Wan, X., and Wang, J. (2025). Planes overhead: How airplane noise impacts home values. NBER Working Paper 34431, National Bureau of Economic Research.

APHA (2021). Noise as a public health hazard. Policy Statement 202115. Adopted October 25, 2021. Accessed September 24, 2026.

Basner, M. and McGuire, S. (2018). WHO environmental noise guidelines for the european region: A systematic review on environmental noise and effects on sleep. *International Journal of Environmental Research and Public Health*, 15(3):519.

Bendtsen, H., Kohler, E., Lu, Q., and Rymer, B. (2010). California–denmark study on acoustic aging of road pavements. *Transportation research record*, 2158(1):122–128.

Bento, A., Freedman, M., and Lang, C. (2015). Who benefits from environmental regulation? evidence from the Clean Air Act Amendments. *Review of Economics and Statistics*, 97(3):610–622.

Blanes, N., Fons, J., Hintzsche, M., Ramos, M. J., Sáinz, M., Milego, R., and Peris, E. (2023). Development of a 2017 baseline to monitor noise under the zero pollution objectives: Methodological document. ETC HE Report 2023/10, European Topic Centre on Human Health and the Environment. Published May 2024.

Board of Audit of Japan (2004). Audit report for fiscal year 2003: Overview of national finances. Technical report, Board of Audit of Japan. Chapter 6, Section 1, Table 2: Expenditure trends for major accounts. In Japanese; title translated.

Board of Audit of Japan (2012). Report on the audit of the implementation of special account reforms. Technical report, Board of Audit of Japan. Appendix Table 1, item 20: Expenditure trends for the Road Improvement Account. In Japanese; title translated.

Boes, S. and Nüesch, S. (2011). Quasi-experimental evidence on the effect of aircraft noise on apartment rents. *Journal of Urban Economics*, 69(2):196–204.

Botta, E. and Yamasaki, S. (2020). Policies, regulatory framework and enforcement for air quality management: The case of Japan. OECD Environment Working Papers 156, OECD Publishing.

Brown, A. L. and van Kamp, I. (2017). WHO environmental noise guidelines for the european region:

A systematic review of transport noise interventions and their impacts on health. *International Journal of Environmental Research and Public Health*, 14(8):873.

BTS (2020). Under a quarter of population exposed to office-type transportation noise. U.S. DOT. November 18. Population exposure estimates for 2018.

BTS (2022). National transportation noise map documentation. Technical report, U.S. DOT. Prepared by the Volpe National Transportation Systems Center.

Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of econometrics*, 225(2):200–230.

CEV (n.d.). EV and Other Next-Generation Vehicle Stock Statistics. Fiscal-year-end stocks; fiscal year 2021 ends in March 2022. In Japanese; title translated. Accessed October 12, 2026.

Chay, K. Y. and Greenstone, M. (2005). Does air quality matter? evidence from the housing market. *Journal of political Economy*, 113(2):376–424.

Currie, J., Davis, L., Greenstone, M., and Walker, R. (2015). Environmental health risks and housing values: evidence from 1,600 toxic plant openings and closings. *American Economic Review*, 105(2):678–709.

Currie, J., Voorheis, J., and Walker, R. (2023). What caused racial disparities in particulate exposure to fall? new evidence from the Clean Air Act and satellite-based measures of air quality. *American Economic Review*, 113(1):71–97.

De Chaisemartin, C. and d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American economic review*, 110(9):2964–2996.

Dean, J. T. (2024). Noise, cognitive function, and worker productivity. *American Economic Journal: Applied Economics*, 16(4):322–360.

EA (2000). Manual for evaluating environmental quality standards for noise, ii: Area evaluation (areas facing roads). Technical report, Government of Japan. Japanese title: Souon ni kakaru kankyou kijun no hyouka manyuaru II: chiiki hyouka hen (douro ni mensuru chiiki). In Japanese. Accessed [date].

EC (2023). Report on the implementation of the environmental noise directive in accordance with article 11 of directive 2002/49/EC. Technical Report COM(2023) 139 final, EC. March 20.

EEA (2025). Environmental noise in europe 2025. EEA Report 05/2025, EEA. Based on the 2022 Environmental Noise Directive reporting round. Online version corrected March 27, 2026.

Environment Agency of Japan (1973). Environmental white paper, 1973. Technical report, Environment Agency of Japan. Overview, Chapter 3, Section 2: The Costs of Environmental Improvement. In Japanese.

EPA (2026). EPA history: Noise and the noise control act. Official agency history web page. Last updated October 19, 2026. Accessed October 27, 2026.

EU (2002). Directive 2002/49/EC relating to the assessment and management of environmental noise. Official Journal of the European Communities, L 189, 18 July 2002, pp. 12–25. European Parliament and Council of the European Union. CELEX 32002L0049.

FHWA (1995). Highway traffic noise analysis and abatement policy and guidance. Technical report, Federal Highway Administration, U.S. Department of Transportation, Washington, D.C. Noise Fundamentals, pp. 3–4.

FHWA (2000). *FHWA Highway Noise Barrier Design Handbook*. Federal Highway Administration.

Report FHWA-EP-00-005. Section 3.5.1.

FHWA (2017). Noise fundamentals. In *Highway Traffic Noise Analysis and Abatement Policy and Guidance*. Federal Highway Administration. Updated August 24, 2017. Available at https://www.fhwa.dot.gov/environment/noise/regulations_and_guidance/polguide/polguide02.cfm.

FHWA (2021). Highway traffic noise barriers at a glance. Federal Highway Administration. Updated October 6, 2021. Available at https://www.fhwa.dot.gov/environment/noise/noise_barriers/design_construction/keepdown.cfm.

FOEN (2025). Speed reductions. Federal Office for the Environment, Switzerland. https://www.bafu.admin.ch/en/speed-reductions. Published November 18, 2025.

Frank, E. G., Auffhammer, M., McLaughlin, D. W., Spiller, B., and Sunding, D. L. (2025). The cost of species protection: The land market impacts of the Endangered Species Act. Working Paper 25-01, Resources for the Future.

Fujimoto, K., Anai, K., and Ohta, R. (2003). Excess attenuation of road traffic noise by detached houses: Predictions when receiving points are lower than houses. *Acoustical Science and Technology*, 24(3):120–129.

Gamper-Rabindran, S. and Timmins, C. (2013). Does cleanup of hazardous waste sites raise housing values? evidence of spatially localized benefits. *Journal of Environmental Economics and Management*, 65(3):345–360.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of econometrics*, 225(2):254–277.

Government of Japan (1968). Noise Regulation Act. https://laws.e-gov.go.jp/law/343AC0000000098. Act No. 98 of 1968. Japanese title: Souon Kisei Hou. In Japanese. Accessed July 14, 2026.

Greenhill, S. (2026). Noise pollution and infant health. Working paper, May 19.

Greenstone, M. (2002). The impacts of environmental regulations on industrial activity: Evidence from the 1970 and 1977 Clean Air Act Amendments and the Census of Manufactures. *Journal of Political Economy*, 110(6):1175–1219.

Greenstone, M. and Gallagher, J. (2008). Does hazardous waste matter? evidence from the housing market and the Superfund program. *Quarterly Journal of Economics*, 123(3):951–1003.

Guski, R., Schreckenberg, D., and Schuemer, R. (2017). WHO environmental noise guidelines for the european region: A systematic review on environmental noise and annoyance. *International Journal of Environmental Research and Public Health*, 14(12):1539.

Hales, D. (2024). Lefant P1 review. Modern Castle. https://moderncastle.com/robot-vacuums/lefant-p1-review/. Updated February 19, 2024.

Han, X., Huang, X., Liang, H., Ma, S., and Gong, J. (2018). Analysis of the relationships between environmental noise and urban morphology. *Environmental Pollution*, 233:755–763.

Interministerial Council (1995). Policy for Implementing Countermeasures in Areas with Severe Road-Traffic Noise. https://www.env.go.jp/hourei/07/000045.html. Interministerial Liaison Council on Road Traffic Pollution Countermeasures. Joint directive of five ministries and agencies, December 1, 1995. In Japanese.

Ito, K., Sallee, J. M., and Halvorson-Fried, C. (2026). Global policy spillovers: How environmental policies propagate through product attributes. NBER Working Paper 35197, National Bureau of Economic Research. September 2, 2026 revision.

Kang, C., Ota, M., and Ushijima, K. (2024). Benefits of diesel emission regulations: Evidence from the world's largest low emission zone. *Journal of Environmental Economics and Management*, 125:102944.

Keiser, D. A., Mazumder, B., Molitor, D., and Shapiro, J. S. (2026). Water works: Causes and consequences of safe drinking water in america. NBER Working Paper 35288, National Bureau of Economic Research.

Keiser, D. A. and Shapiro, J. S. (2019). Consequences of the clean water act and the demand for water quality. *Quarterly Journal of Economics*, 134(1):349–396.

Kephalopoulos, S., Paviotti, M., and Anfosso-Lédée, F. (2012). Common noise assessment methods in Europe (CNOSSOS-EU). Technical Report EUR 25379 EN, European Commission, Joint Research Centre, Luxembourg.

Kochi, I., Matsuoka, S., Memon, M. A., and Shirakawa, H. (2002). Cost benefit analysis of the sulfur dioxide emissions control policy in Japan. *Environmental Economics and Policy Studies*, 4(4):219–233.

Lindgren, S. (2021). A sound investment? traffic noise mitigation and property values. *Journal of Environmental Economics and Policy*, 10(4):428–445.

Matsuno, Y. and Ueta, K. (2002). Environmental policies of local governments in Japan focusing on pollution control agreements: Report of a questionnaire survey. *Keizai Ronso Bessatsu: Chosa to Kenkyu*, 23:1–155. In Japanese.

Menge, C. W., Rossano, C. F., Anderson, G. S., and Bajdek, C. J. (1998). FHWA traffic noise model, version 1.0: Technical manual. Technical Report FHWA-PD-96-010; DOT-VNTSC-FHWA-98-2, Federal Highway Administration, U.S. Department of Transportation.

Mikhailenko, P., Piao, Z., Kakar, M. R., Bueno, M., Athari, S., Pieren, R., Heutschi, K., and Poulikakos, L. (2022). Low-noise pavement technologies and evaluation techniques: A literature review. *International Journal of Pavement Engineering*, 23(6):1911–1934.

Ministry of Land, Infrastructure and Transport, Japan (2007). Overview of the road budget for fiscal year 2007. Technical report, Road Bureau and City and Regional Development Bureau, Ministry of Land, Infrastructure and Transport. Section on roadside environmental conservation and improvement. In Japanese; title translated.

Ministry of the Environment, Japan (2001). Environmental white paper 2001. Technical report, Ministry of the Environment, Japan. Transport and traffic, subsection on road structure and roadside environmental improvements. In Japanese; title translated.

MLIT (2000–2010). Public notice of land prices. Available at https://www.mlit.go.jp/totikensangyo/totikensangyo_fr4_000444.html.

MLIT (2002–2009). Motor vehicle transport survey. Annual reports. Available at https://www.mlit.go.jp/k-toukei/jidousya202003.html.

MLIT (2003). *Road Administration Performance Plan for Fiscal Year 2003: Roadside Noise*. Ministry of Land, Infrastructure, Transport and Tourism, Japan. Section 15. In Japanese; title translated.

MLIT (2004). *Road Environmental Census: Survey Manual (Summary Version)*. Ministry of Land, Infrastructure, Transport and Tourism. Fiscal year 2004. In Japanese; title translated. Available at https://www.cgr.mlit.go.jp/chiki/doyroj/ir/souon/h16_tyousagaiyou.pdf.

MLIT (2005–2021). Road traffic census: General traffic volume survey. Available at https://www.

mlit.go.jp/statistics/details/sample03_2_00044.html.

MLIT (2010–2022). Motor vehicle fuel consumption survey. Annual reports. Available at https://www.mlit.go.jp/k-toukei/nenryousyouhiryou.html.

MOE (1998). Environmental Quality Standards for Noise. https://www.env.go.jp/kijun/oto1-1.html. Environment Agency Notification No. 64, September 30, 1998; effective April 1, 1999. In Japanese. Accessed July 14, 2026.

MOE (2022). Continuous monitoring of road traffic noise results. Data cover 2002–2022. Disseminated through the National Institute for Environmental Studies Environmental GIS. Available at https://tenbou.nies.go.jp/gisplus/download.html.

MOE (n.d.a). Historical motor-vehicle noise limits. Japanese table: Jidosha tantai soon kiseichi. Title translated. Accessed October 5, 2026.

MOE (n.d.b). Overview of the Noise Regulation Act. https://www.env.go.jp/air/noise/low-gaiyo.html. In Japanese. Accessed July 14, 2026.

MOE (n.d.c). Vehicle production by noise-regulation category. Reference 4. Covers January–December 1997. Environment Agency survey; in Japanese; title translated.

Moretti, E. and Wheeler, H. (2025). The traffic noise externality: Costs, incidence and policy implications. Technical report, National Bureau of Economic Research.

National Academy of Engineering (2010). *Technology for a Quieter America*. The National Academies Press, Washington, DC.

National Research Council (1977). Noise abatement: Policy alternatives for transportation. Analytical Studies for the U.S. Environmental Protection Agency VIII, National Academy of Sciences, Washington, D.C.

NHTSA (2016). Minimum sound requirements for hybrid and electric vehicles: Final environmental assessment. Technical Report DOT HS 812 347, NHTSA.

NIES (2025). Environmental gis: Air-pollution monitoring data. Available at https://tenbou.nies.go.jp/gis/explain/atm/moni.html.

Nishitateno, S. and Burke, P. J. (2021). Willingness to pay for clean air: Evidence from diesel vehicle registration restrictions in japan. *Regional Science and Urban Economics*, 88:103657.

Okada, Y., Fukushima, A., Yamauchi, K., and Sakamoto, S. (2019). Road traffic noise prediction model "ASJ RTN-Model 2018" proposed by the acoustical society of Japan: Part 2, calculation model of sound emission of road vehicles. In *Proceedings of the 23rd International Congress on Acoustics*, pages 3696–3703, Aachen, Germany.

Okada, Y., Ichikawa, T., Yoshihisa, K., Uemura, T., Yonemura, M., Lee, H., Sakamoto, S., and Yamauchi, K. (2020). Sound power level of road vehicles running on dense asphalt pavement at various sites in Japan. *Acoustical Science and Technology*, 41(3):598–606.

Ortiz-Villavicencio, M. and Sant'Anna, P. H. (2025). Better understanding triple differences estimators. *arXiv preprint arXiv:2505.09942*.

Ragettli, M. S., Goudreau, S., Plante, C., Perron, S., Fournier, M., and Smargiassi, A. (2016). Annoyance from road traffic, trains, airplanes and from total environmental noise levels. *International journal of environmental research and public health*, 13(1):90.

Rasmussen, R. O., Bernhard, R. J., Sandberg, U., and Mun, E. P. (2007). The little book of quieter pavements. Technical Report FHWA-IF-08-004, FHWA.

Sakamoto, S., Okada, Y., Fukushima, A., Matsumoto, T., and Tajika, T. (2019). Road traffic noise prediction model "ASJ RTN-Model 2018" proposed by the acoustical society of Japan: Part 1, outline of the calculation model. In *Proceedings of the 23rd International Congress on Acoustics*, pages 3690–3695, Aachen, Germany.

Sakamoto, S., Xu, W., Fukuda, T., and Yonemura, M. (2023). Validation of calculation method of road traffic noise behind building complex in ASJ RTN-Model 2018 by field measurements. In *INTER-NOISE and NOISE-CON Congress and Conference Proceedings*, volume 265. INTER-NOISE 2023, Chiba, Japan.

Sandberg, U. (2001). Noise emissions of road vehicles: Effect of regulations. Final Report 01-1, International Institute of Noise Control Engineering.

SBJ (2000). 2000 population census: Small-area tabulations. Tables on age composition and educational attainment. Available at https://www.stat.go.jp/data/kokusei/2000/kohyo.html.

SBJ (2026). Japanese Statistical Yearbook 2026: Motor Vehicles Owned by Kind. Table 13-5. End-of-fiscal-year vehicle stocks; fiscal year 2021 ends in March 2022.

Schlauch, G. and Greenhill, S. (2026). Data centers lower nearby housing values. Working Paper 366, Energy Institute at Haas, University of California, Berkeley.

Shkembi, A., Patel, K., Smith, L. M., Meier, H. C. S., and Neitzel, R. L. (2026). Racial and ethnic inequities to noise pollution from transportation- and work-related sources in the United States. *Journal of Exposure Science & Environmental Epidemiology*, 36:211–220. First published online July 17, 2025.

Sueoka, S., Uchida, H., Kikuchi, H., Kamoshida, H., Kadoya, M., and Tanaka, S. (2009). An investigation of the new index of the noise level in Japan. *Journal of Environmental Laboratories Association*, 34(4):254–261. In Japanese. https://tenbou.nies.go.jp/science/institute/region/journal/JELA_3404022_2009.pdf.

Sugasawa, T., Kuroda, Y., Nomura, K., Yasuda, S., and Yoshida, J. (2025). Urban responses to additional aircraft noise: Evidence from a natural experiment in Tokyo. Working paper, SSRN 5893696.

Supreme Court of Japan (1995). National Route 43 Pollution Litigation Judgment. Second Petty Bench, July 7, 1995; Minshu 49(7):1870 and 2599. In Japanese.

Toyota (1997). Toyota Launches the Revolutionary PRIUS Hybrid Passenger Vehicle. Press release, October 14. Announces sales in Japan beginning December 10, 1997.

U.S. Congress (1972). Noise control act of 1972. Public Law 92-574, 86 Stat. 1234; codified as amended at 42 U.S.C. 4901 et seq.

U.S. Congress (1978). Quiet communities act of 1978. Public Law 95-609, 92 Stat. 3079.

U.S. DOT (2025). Benefit-cost analysis guidance for discretionary grant programs. Technical report, U.S. Department of Transportation, Office of the Secretary. Noise pollution discussion, p. 26.

U.S. EPA (2011). The benefits and costs of the Clean Air Act from 1990 to 2020. Final report, revision a, U.S. Environmental Protection Agency.

U.S. EPA (2024). EPA's final Lead and Copper Rule Improvements: Technical fact sheet: Summary of benefits and costs. Fact Sheet EPA 816-F-24-008, U.S. Environmental Protection Agency, Office of Water.

van Kempen, E., Casas, M., Pershagen, G., and Foraster, M. (2018). WHO environmental noise

guidelines for the european region: A systematic review on environmental noise and cardiovascular and metabolic effects: A summary. *International Journal of Environmental Research and Public Health*, 15(2):379.

VanderWerp, D. (2024). Latest Tesla Model 3 is vastly quieter than before, with far more highway range. Car and Driver. https://www.caranddriver.com/news/a62595445/2024-tesla-model-3-quieter-more-highway-range-tested/. Published October 14, 2024.

von Graevenitz, K. (2018). The amenity cost of road noise. *Journal of Environmental Economics and Management*, 90:1–22.

Walker, W. R. (2013). The transitional costs of sectoral reallocation: Evidence from the Clean Air Act and the workforce. *Quarterly Journal of Economics*, 128(4):1787–1835.

WHO (2018). *Environmental Noise Guidelines for the European Region*. WHO Regional Office for Europe, Copenhagen.

Wilk, J., Szyszlak-Bargłowicz, J., Słowik, T., Zając, G., and Wasilewski, J. (2025). Assessment of acoustic climate and traffic noise pollution: A case study. *Journal of Ecological Engineering*, 26(4):275–284.

Winke, T. (2017). The impact of aircraft noise on apartment prices: A differences-in-differences hedonic approach for Frankfurt, Germany. *Journal of Economic Geography*, 17(6):1283–1300.

Xu, C., Zhang, Y., Cheng, B., Li, L., and Zhang, M. (2020). Study on environmental Kuznets curve for noise pollution: A case of 111 Chinese cities. *Sustainable Cities and Society*, 63:102493.

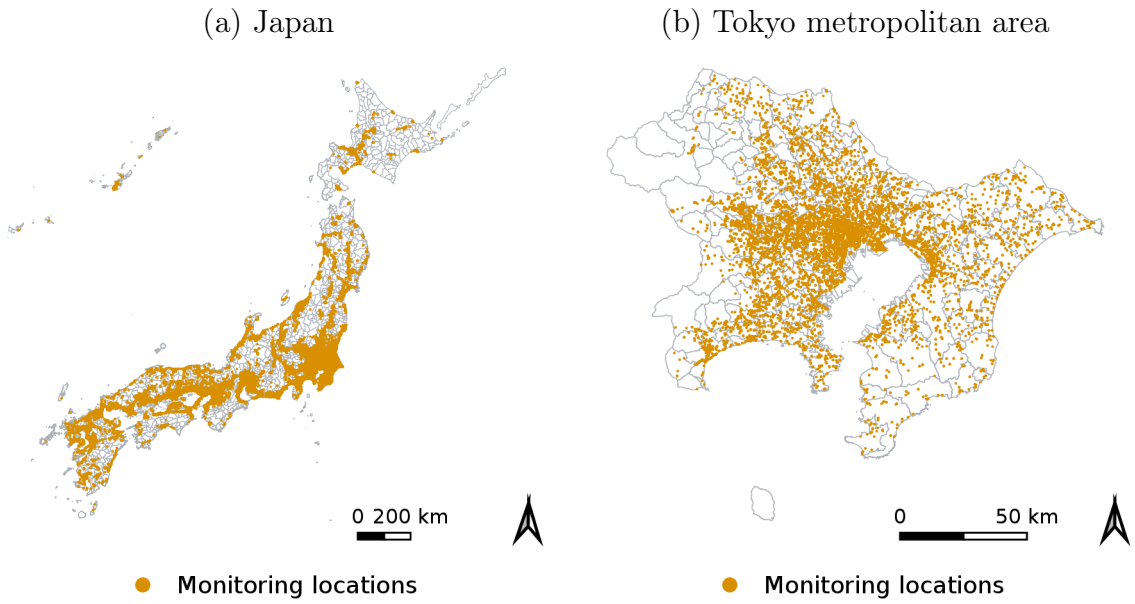
Xu, G., Li, K., Li, C., Wang, H., Leng, Z., and Chen, X. (2024). Noise reduction performance and maintenance time of porous asphalt pavement. *Construction and Building Materials*, 452:138913.

Yamauchi, K. (2025). A-weighted sound power level of electric vehicles in accelerating and decelerating conditions. *Acoustical Science and Technology*, 46(4):494–500.

Zenkeijikyo (2025). Kei-vehicle statistics. Available at https://www.zenkeijikyo.or.jp/statistics.

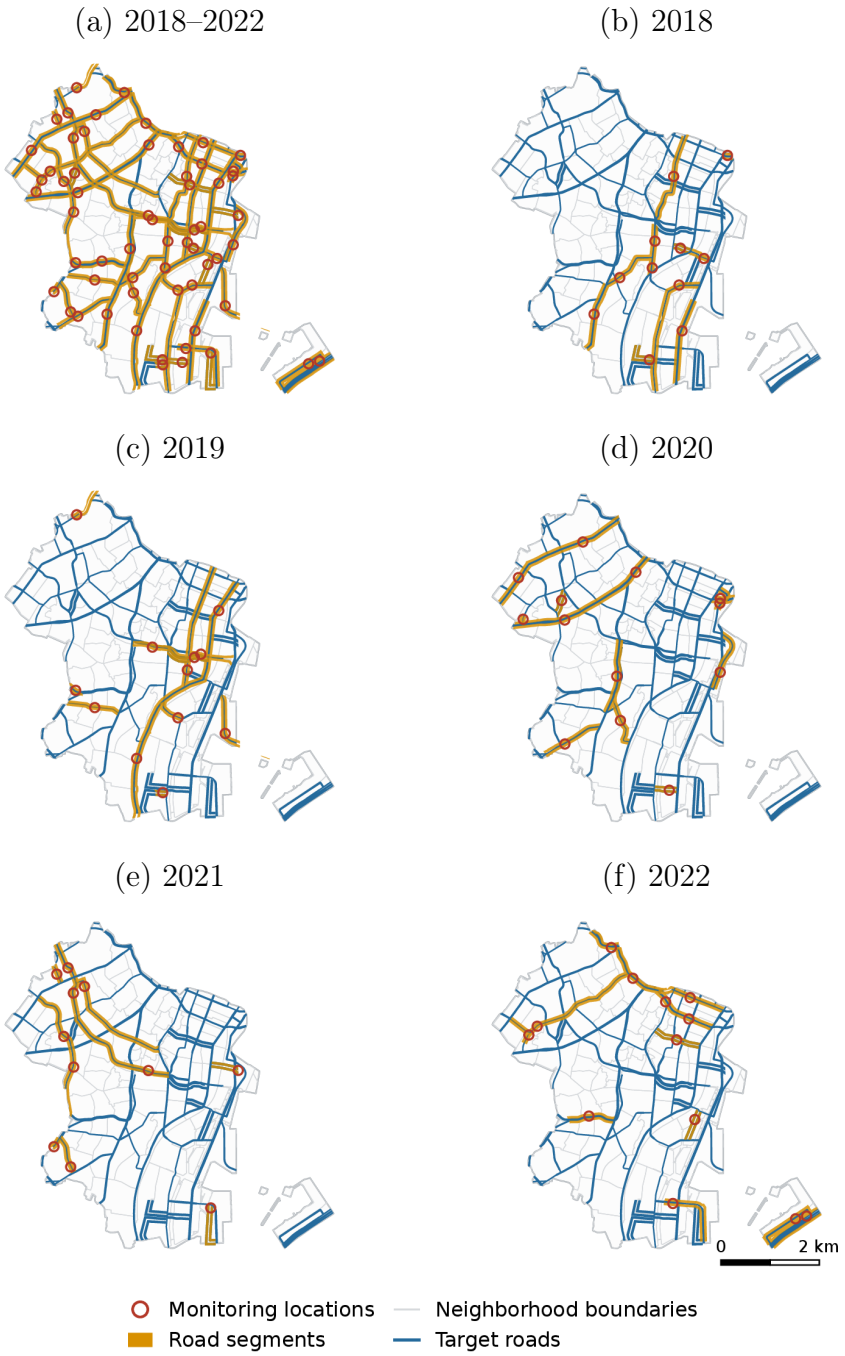
Figures

Figure 1: Coverage of the Continuous Monitoring of Road Traffic Noise



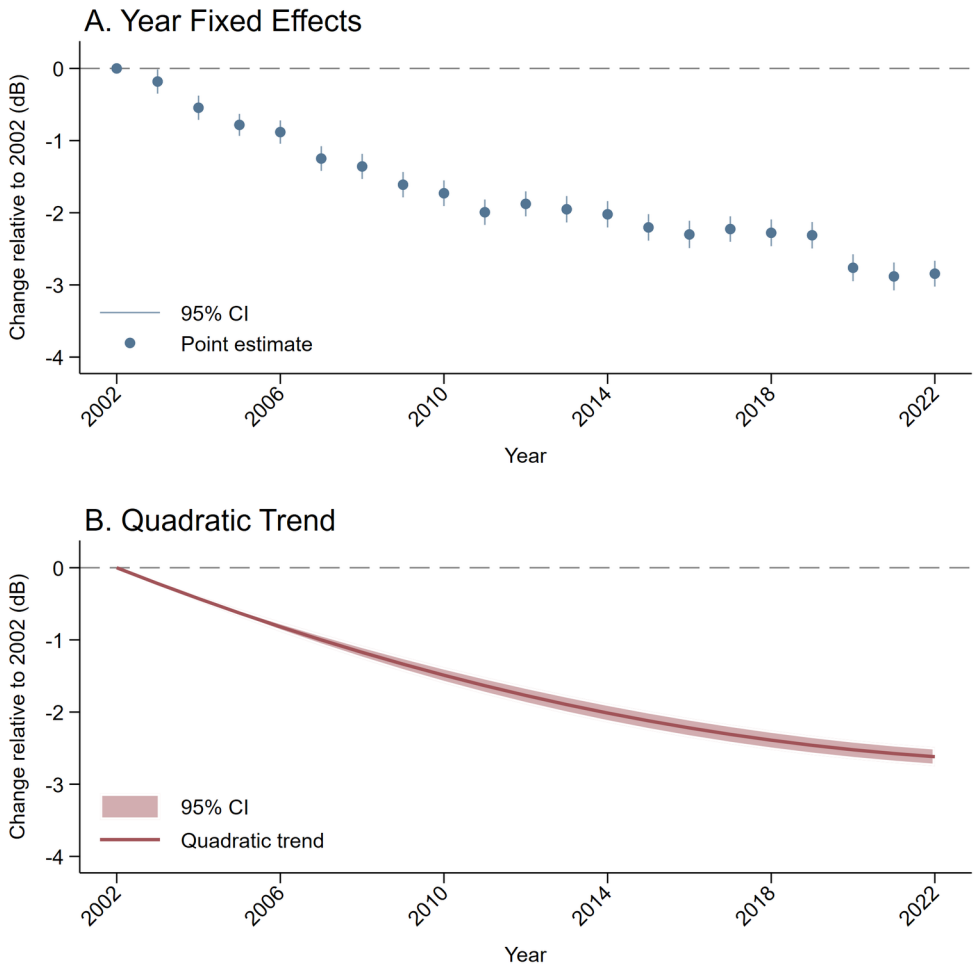
Notes: Both panels show monitoring locations observed during 2002–2022. Panel A covers Japan. Panel B covers Tokyo, Saitama, Chiba, and Kanagawa. Gray lines show administrative boundaries.

Figure 2: Rotating Road-Traffic Noise Monitoring in Minato Ward, 2018–2022



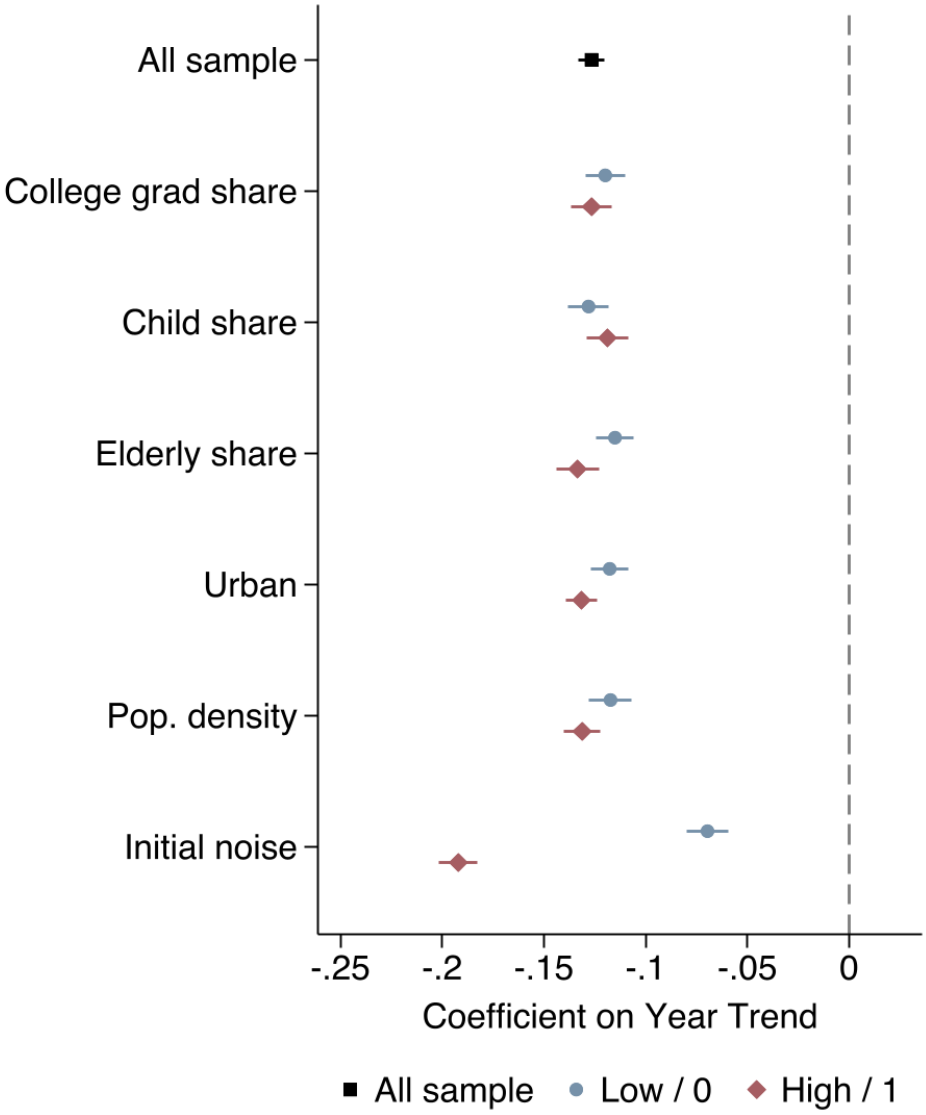
Notes: Panel A pools observations from 2018–2022; Panels B–F show each year. Road segments retain the original CMRTN evaluation-area polygons. Target roads have at least two lanes for expressways, national roads, and prefectural roads, or four lanes for municipal roads. The target layer uses the 2021 Road Traffic Census and qualifying CMRTN ward-road records and stays fixed across panels. It excludes discretionary municipal additions. Municipal-road coverage may be incomplete, and roadside building use is not screened. Sources: CMRTN, Road Traffic Census, and Population Census.

Figure 3: Trends in 24-hour Road Traffic Noise from 2002 to 2022



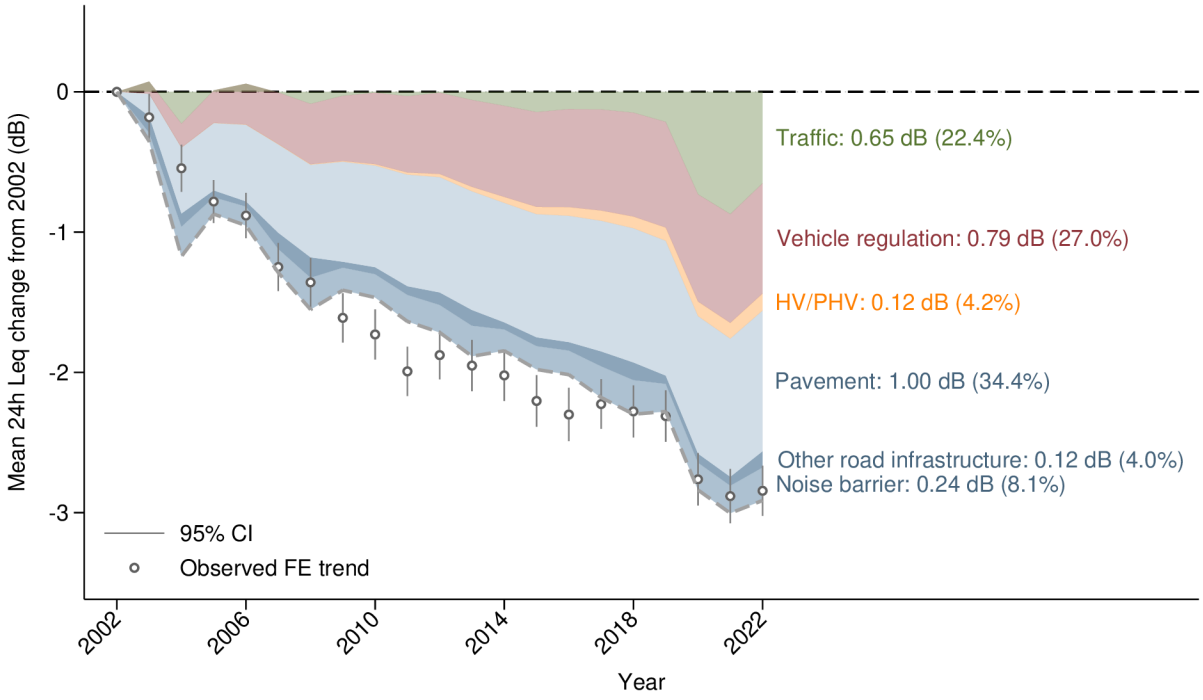
Notes: This figure plots trends in the 24-hour equivalent continuous sound level. Panel A plots year-specific coefficients from a specification with year fixed effects, with 2002 as the omitted reference year. Vertical bars represent 95 percent confidence intervals. Panel B plots the predicted quadratic time trend, normalized to zero in 2002. The shaded area represents the pointwise 95 percent confidence interval for the quadratic prediction. All specifications include the full set of controls and road-segment fixed effects. Standard errors are clustered at the road-segment level.

Figure 4: Heterogeneous Trends in Noise Pollution



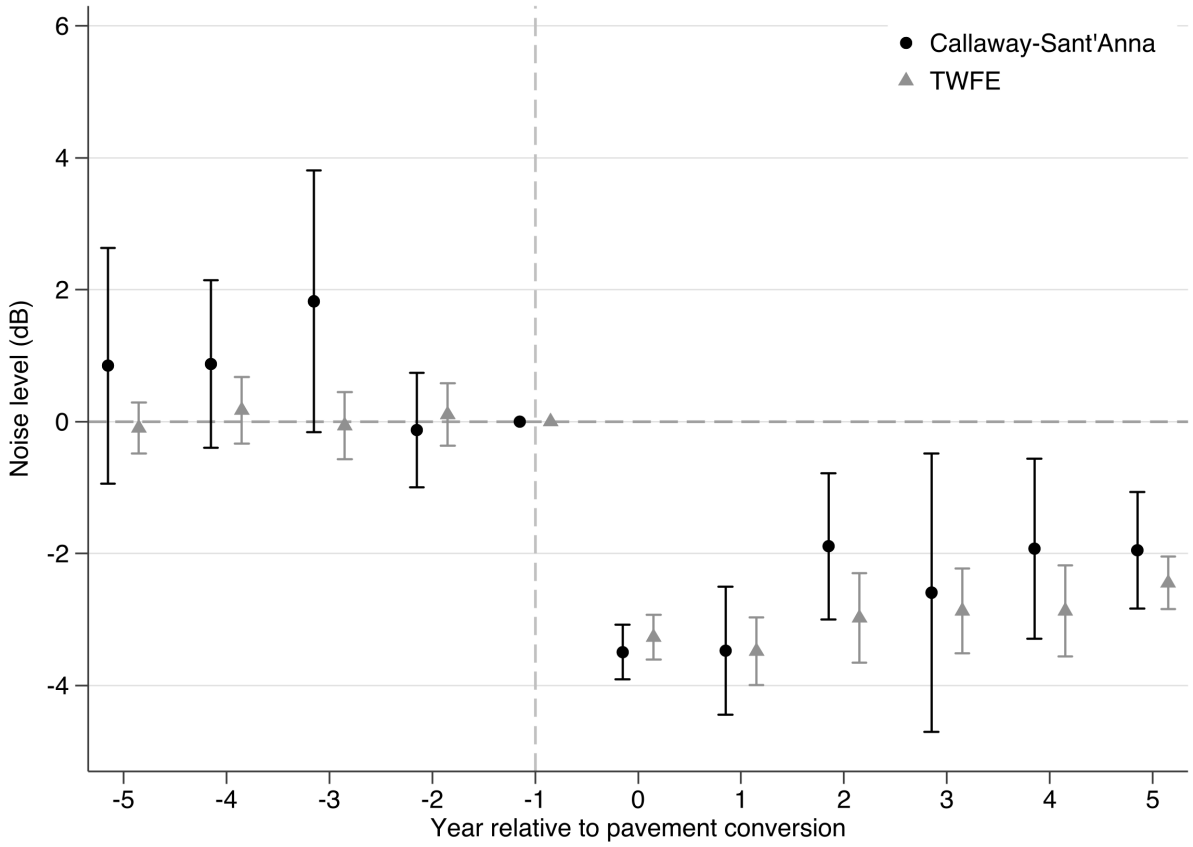
Notes: This figure plots heterogeneous trends in 24-hour equivalent noise levels across groups. Groups are defined using median splits, and urban areas are defined as road segments located in the three major metropolitan areas (Tokyo, Chubu, and Kansai). The estimates are based on coefficients on the linear year trend. The specification includes all controls and road-segment fixed effects. Standard errors are clustered at the road-segment level. Vertical bars represent 95 percent confidence intervals.

Figure 5: National Engineering Decomposition of 24-Hour Noise Change



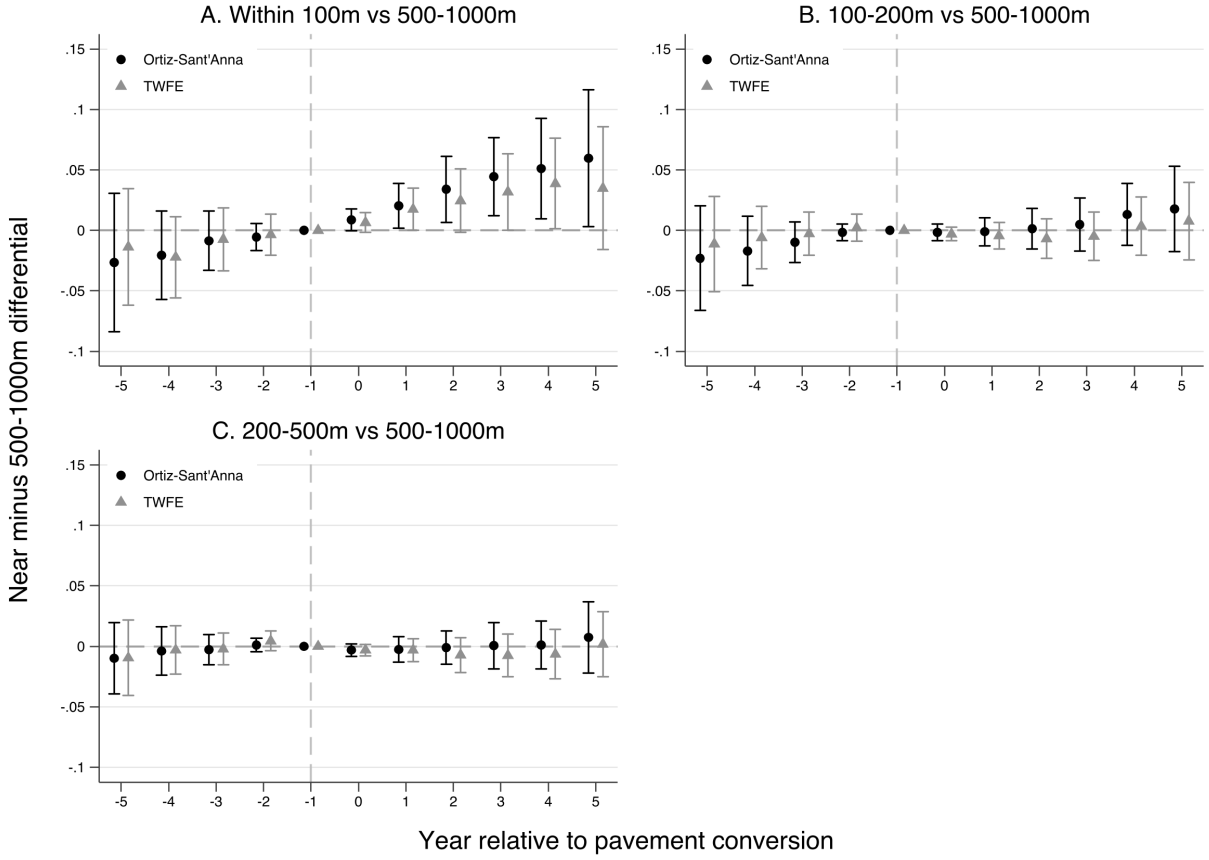
Notes: Circles show the regression estimates, and bars show 95% confidence intervals. The dashed line shows the model prediction. Shaded areas show each factor's contribution to the change since 2002. Labels report the 2022 contributions in dB and as shares of the predicted decline.

Figure 6: Effect of Pavement Conversion on Noise Levels



Notes: Circles show Callaway–Sant'Anna estimates; triangles show TWFE. Both series control for road width, lane count, and the log distance of the monitoring point from the road center. The outcome is the 24-hour equivalent noise level, and event time is measured in survey years. Bars show 95% pointwise confidence intervals with road-segment clustering. CS uses never-treated comparisons; TWFE includes segment and survey-year fixed effects, takes year -1 as the omitted reference, and pools years at or beyond -5 and 5 into the endpoint coefficients. Table 5 reports the average over years 0–5 and describes the controls and sample.

Figure 7: DDD: Effects of Pavement Conversion on Land Prices



Notes: Each panel compares the indicated distance band (0–100m, 100–200m, or 200–500m) with the 500–1000m band. Circles show Ortiz-Villavicencio–Sant'Anna estimates; triangles show TWFE. Both series control for road width and lane count; TWFE interacts them with band and year. The outcome is log land price. Bars show 95% pointwise confidence intervals with road-segment clustering. OVS uses never-treated comparisons and 999 clustered multiplier bootstrap draws. For TWFE, year −1 is the omitted reference. OVS reports exact event years; the TWFE endpoint coefficients pool years at or beyond −5 and 5. Table 5 reports the averages over years 0–5 and describes the controls and sample.

Tables

Table 1: Socioeconomic Characteristics and Road Traffic Noise

	Dependent variable: avg. Leq_{24h}, 2002–2006				
	(1)	(2)	(3)	(4)	(5)
College graduate share	0.63		-0.53	-1.29	-1.70
	(0.90)		(0.97)	(1.16)	(1.05)
Share age 0–14		-6.67***	-7.00***	-5.77***	-6.96***
		(2.11)	(2.20)	(2.16)	(2.23)
Share age 65+		-7.03***	-7.27***	-6.58***	-6.92***
		(1.27)	(1.39)	(1.24)	(1.19)
Population density				30.9**	27.1*
				(15.4)	(14.6)
Urban dummy				0.07	2.08***
				(0.16)	(0.65)
Observations	2,855	2,836	2,830	2,830	2,830
R-squared	0.064	0.079	0.079	0.081	0.120
Controls	Yes	Yes	Yes	Yes	Yes
Prefecture FE	No	No	No	No	Yes

Notes: This table reports OLS estimates of the association between socioeconomic characteristics and baseline road traffic noise. The dependent variable is the road-segment average 24-hour equivalent continuous sound level over 2002–2006. Controls include the log distance between the monitoring point and the matched road segment, ground-height-bin fixed effects, and measurement-month fixed effects. Standard errors are clustered by municipality and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: Trends in Road Traffic Noise

	Noise level			Exceedance rate	
	24h	Daytime	Nighttime	Daytime	Nighttime
	(1)	(2)	(3)	(4)	(5)
Panel A. Linear trend					
Year	-0.13^{***}	-0.12^{***}	-0.18^{***}	-0.013^{***}	-0.014^{***}
	(0.0032)	(0.0031)	(0.0037)	(0.00046)	(0.00044)
Observations	43,369	43,370	43,369	43,370	43,370
R-squared	0.814	0.807	0.834	0.639	0.697
Panel B. Quadratic trend					
Year	-0.22^{***}	-0.20^{***}	-0.31^{***}	-0.028^{***}	-0.028^{***}
	(0.010)	(0.0100)	(0.012)	(0.0015)	(0.0014)
Year^2	0.0046^{***}	0.0041^{***}	0.0060^{***}	0.00072^{***}	0.00063^{***}
	(0.00042)	(0.00042)	(0.00049)	(0.000062)	(0.000058)
Observations	43,369	43,370	43,369	43,370	43,370
R-squared	0.815	0.808	0.835	0.642	0.698

Notes: This table reports OLS estimates of time trends in road traffic noise. Panel A reports specifications with a linear year trend. Panel B reports specifications with both linear and quadratic year trends. The dependent variables are the 24-hour equivalent continuous sound level, daytime noise, nighttime noise, and indicators for whether daytime and nighttime noise exceed the relevant environmental standard. Standard errors clustered at the road-segment level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Parameters Used in the Noise Decomposition

Mechanism	Parameter (1)	Value (2)	Interpretation (3)
Panel A. Traffic conditions			
Traffic volume	α_Q	1.0	Acoustic-energy elasticity with respect to traffic volume
Traffic speed	γ	1.0	Acoustic-energy elasticity with respect to speed
Large-vehicle share	δ_L	7.4 dB	Large-vehicle noise premium relative to small vehicles
Panel B. HV/PHV adoption			
HV/PHV vehicles	d_H	1.5 dB	Noise reduction relative to comparable conventional vehicles
Panel C. Vehicle noise standards			
Small vehicles	d_s	2.0 dB	Tightening of light-vehicle acceleration-noise limits
Large vehicles	d_L	2.6 dB	Tightening of heavy-vehicle acceleration-noise limits
Road-level pass-through	θ_R	0.5	Assumed multiplier on the single-vehicle dB reduction
Panel D. Road infrastructure			
Drainage pavement	d_P	2.5 dB	Expected noise reduction
Environmental facility belt	d_E	6.0 dB	Expected noise reduction
Noise barrier	d_B	5.0 dB	Expected noise reduction
Sound-absorbing panel	d_A	3.0 dB	Expected noise reduction

Notes: HV and PHV denote conventional hybrids and plug-in hybrids; dB denotes decibels.

Table 4: Decomposition of Changes in Road Traffic Noise

Decomposition term	2002–2012		2013–2022		2002–2022	
	Change (dB) (1)	Share (%) (2)	Change (dB) (3)	Share (%) (4)	Change (dB) (5)	Share (%) (6)
Panel A. Traffic conditions						
Total	−0.01	0.35%	−0.65	54%	−0.65	22%
Traffic volume	−0.12	7.2%	−0.19	16%	−0.31	11%
Speed	0.30	−18%	−0.37	31%	−0.07	2.5%
Heavy-vehicle share	−0.18	11%	−0.08	6.8%	−0.27	9.1%
Panel B. HV/PHV adoption						
Total	−0.02	1.3%	−0.10	8.3%	−0.12	4.2%
Panel C. Vehicle noise standards						
Total	−0.58	34%	−0.21	17%	−0.79	27%
Panel D. Road infrastructure						
Total	−1.11	65%	−0.25	21%	−1.35	46%
Pavement	−0.82	48%	−0.18	15%	−1.00	34%
Noise barriers	−0.20	11%	−0.04	3.3%	−0.24	8.1%
Sound absorbers	−0.02	1.4%	−0.01	0.54%	−0.03	1.1%
Buffer zones	−0.06	3.8%	−0.02	1.7%	−0.08	2.9%
Predicted change	−1.72	100%	−1.20	100%	−2.91	100%
Year-FE trend	−1.88		−0.97		−2.84	

Notes: Negative dB values indicate noise reductions. Percentages show each factor's share of the predicted decline in that period; negative shares indicate contributions that offset the decline. The 2013–2022 columns show additional changes after 2012. The Year-FE trend shows the regression estimate. Rounding can cause small differences in totals.

Table 5: Effects of Pavement Conversion on Noise Levels and Land Prices

	Noise level (dB)	Log land prices		
		Within 100m	100–200m	200–500m
	(1)	(2)	(3)	(4)
Panel A: Callaway–Sant'Anna				
Pavement conversion	−2.55***	—	—	—
	(0.32)			
Panel B: Ortiz-Villavicencio–Sant'Anna				
Pavement conversion × Near band	—	0.036**	0.006	0.000
		(0.015)	(0.009)	(0.008)
Panel C: Two-way fixed effects				
Pavement conversion	−3.14***	—	—	—
	(0.10)			
Pavement conversion × Near band	—	0.030*	0.000	−0.003
		(0.016)	(0.011)	(0.009)
Road structure controls	Yes	Yes	Yes	Yes
Measurement controls	Yes	No	No	No
Observations	18,100	39,930	45,639	62,194
Clusters (segments)	10,628	2,196	2,433	2,930

Notes: Column (1) reports effects on the 24-hour equivalent noise level in decibels. Columns (2)–(4) report triple-difference effects on log land prices, comparing the indicated distance band with the 500–1000m band. Pavement conversion indicates resurfacing from dense-graded to low-noise pavement, and near band equals one for parcels in the indicated band. Dashes mark estimates not reported for that outcome. Panels A and B use never-treated comparisons and report the average of the event-time effects over years 0–5; Panel C reports pooled post-treatment coefficients. Road structure controls are road width and lane count. The measurement control is the log distance of the monitoring point from the road center. Land-price controls use each segment's earliest observed values. The noise regressions include segment and survey-year fixed effects. The land-price specification in Panel C includes parcel fixed effects, band-by-year fixed effects, a common post-conversion term, and controls interacted with band and year. Observations are station-survey years in Column (1) and parcel-years, including the comparison band, in Columns (2)–(4). Standard errors, in parentheses, are clustered by road segment; Panel B uses 999 multiplier bootstrap draws. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Capitalized benefit of the policy-driven decline in road traffic noise

	Value
WTP per decibel reduction	1.42 % of land value
Decline attributed to policy	2.14 dB
Residential land stock	222.5 trillion yen
Capitalized benefit	**6.8 trillion yen**
per household	438,000 yen
vehicle emission standards (0.79 dB)	2.5 trillion yen
low-noise pavement (1.00 dB)	3.2 trillion yen
noise barriers and other infrastructure (0.35 dB)	1.1 trillion yen

Notes: The WTP per decibel is the land-price effect within 100m divided by the noise effect, both averaged over years 0 to 5 after conversion (Table 5). The decline attributed to policy is the sum of the 2022 contributions of the policy channels in the engineering model (Section 5). The residential land stock is the value of residential land within 100m of the roads where the Ministry of the Environment evaluated noise in 2002 to 2023 (Appendix E). The capitalized benefit is the WTP per decibel times the decline times the stock, split across channels in proportion to their decibel contributions. The per-household figure divides the capitalized benefit by the 15.5 million households within 100m of these roads.

ONLINE APPENDIX

Combating Noise Pollution: Policy, Exposure, and Welfare

Takeru Sugasawa HRAFJ
Shinsaku Takikawa LSE
Takanao Tanaka UC Berkeley

A Data Construction and Spatial Linkage

A-1 CMRTN

A-1.1 Panel Construction

We construct the noise panel using CMRTN data from the National Institute for Environmental Studies (NIES, 2023). The data do not provide IDs that link measurement locations to road segments or allow us to track the same road segments across years. We therefore first match each measurement location to the nearest road segment within the same municipality and year, retaining only matches within 200 meters. We then identify overlapping road segments across years within each municipality and assign them a common ID. The resulting panel follows road segments over time, allowing measurement locations within each segment to change.

A-1.2 Measurement Procedures

MOE guidelines standardize measurement locations, microphone placement, and equipment. Local governments normally select a point at the road boundary with a clear view of a straight, relatively flat road section, away from intersections and other major noise sources. Technicians place microphones between 1.2 and 5.0 meters above ground. If they cannot measure at the road boundary, they choose another location that provides a clear view of the road and limits sound reflection. Local governments may change measurement locations across years as road conditions and the surrounding area change. Technicians use certified sound-level meters that meet Japanese Industrial Standards (JIS) and check the instruments before taking measurements.

The guidelines also specify measurement timing and duration. Local governments generally take measurements on weekdays in autumn, excluding holidays. They may combine measurements from several dates to better represent annual conditions. When traffic is frequent and stable, the guidelines allow shorter measurements lasting at least 10 minutes in each hour. When traffic is sparse or irregular, authorities use longer samples, continuous monitoring, or measurements of individual vehicles. Authorities remove recording periods with unrelated sounds before calculating noise levels (EA, 2000; MOE, 2005, 2011, 2015).

A-2 Road Environmental Census Panel Construction

The Road Environmental Census provides addresses, road names, and road-segment lengths. We first locate each reported address on a map and find the nearest point on the named road. We then draw a circle around this point, with a radius equal to half the reported

length, and keep the part of the road inside the circle. Finally, we link these segments to land-price sites and air-pollution monitoring stations.

A-3 Other Data

A-3.1 Trend Analysis

For the trend analysis in Section 4, we use baseline neighborhood characteristics from small-area tabulations of the 2000 Population Census (SBJ, 2000). We define the college-graduate share as the number of university and graduate-school graduates divided by the total number of graduates reported in the educational-attainment table. Using population counts in five-year age groups, we construct the share of children aged 0–14 and the share of residents aged 65 or older. We calculate population density as the small area's total population divided by its land area. We match these characteristics to noise-monitoring locations using municipality and small-area identifiers and hold them fixed at their 2000 values throughout the trend analysis.

A-3.2 Engineering Analysis

For the engineering analysis in Section 5, we obtain traffic volume, travel speed, and heavy-vehicle shares from the 2005, 2010, 2015, and 2021 Road Traffic Census (MLIT, 2021b).[1] The Census covers major roads, using different road segments from the CMRTN. We obtain annual vehicle mileage from MLIT's Motor Vehicle Transport Survey for 2002–2009 and Motor Vehicle Fuel Consumption Survey for 2010–2022 (MLIT, 2009b, 2022). These annual data help us capture changes in traffic volume between census years.

AIRIA provides annual national stocks of hybrid and plug-in hybrid passenger vehicles and the total passenger-vehicle stock for 2002–2022 (AIRIA, 2025a,b). Zenkeijikyo provides annual new light-vehicle sales and existing light-vehicle stocks, which we use to estimate vehicle replacement (Zenkeijikyo, 2025). Appendix Section C-1 explains how we construct the model inputs from these sources.

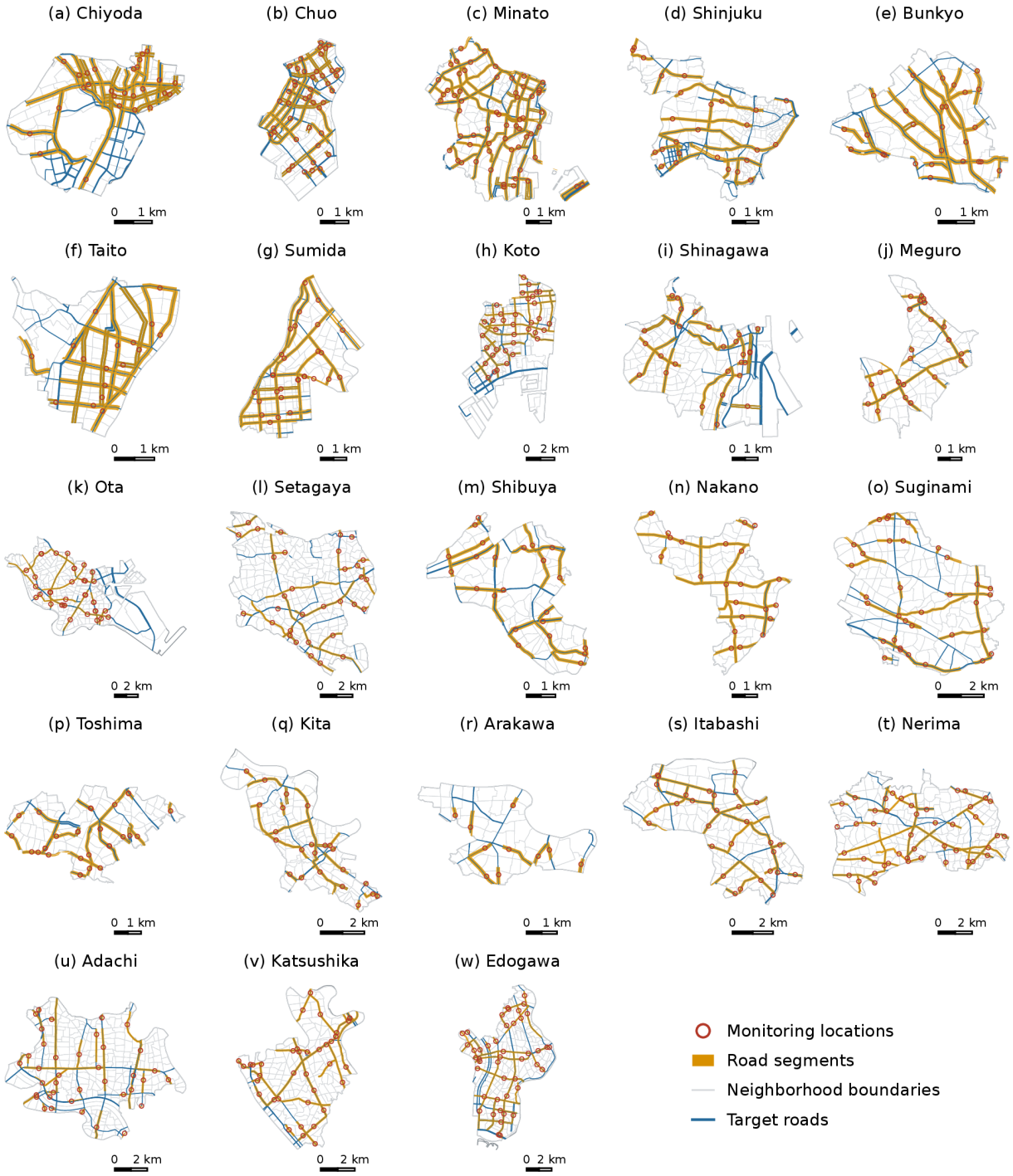
A-3.3 Willingness to Pay

For the willingness-to-pay analysis in Section 6, we use roadside air pollution measurements to examine whether pavement improvements also affect air quality. We obtain annual monitoring-station observations from the National Institute for Environmental Studies' Environmental GIS (NIES, 2025). We use annual mean concentrations of nitrogen oxides (NOx),

[1] MLIT postponed the census scheduled for 2020 until 2021 because of the COVID-19 pandemic (MLIT, 2021a).

suspended particulate matter (SPM), and carbon monoxide (CO) from roadside automobile-exhaust monitoring stations during 2000–2010. We measure NOx and CO in parts per million and SPM in milligrams per cubic meter. We match each station to the nearest Road Environmental Census segment. Note that only a small share of the road segments in our land-price sample have air pollution data, including few segments that received pavement upgrades.

Figure A1: Road-Traffic Noise Monitoring in Tokyo's 23 Wards, 2018–2022



Notes: Each panel pools monitoring locations and road segments observed during 2018–2022 in one ward. Road segments retain the original CMRTN evaluation-area polygons. Target roads follow the road-class and lane-count rules in Figure 2, including qualifying roads without monitoring observations in these years. Municipal-road coverage may be incomplete, and roadside building use is not screened. Each panel has its own scale bar. Sources: CMRTN, the 2021 Road Traffic Census, and Population Census neighborhood boundaries (2010, except 2000 for Chiyoda).

Table A1: Summary Statistics

Variable	Mean	SD	N
Panel A. CMRTN			
Noise exposure			
24-hour noise level (dB)	66.4	4.50	46,543
Daytime noise level (dB)	67.4	4.38	46,544
Nighttime noise level (dB)	62.5	5.79	46,543
Daytime environmental-standard exceedance (%)	33.0	47.0	46,544
Nighttime environmental-standard exceedance (%)	38.9	48.8	46,544
Demographic and neighborhood characteristics			
College-graduate share (%)	21.3	10.2	34,830
Share aged 0–14 (%)	13.9	4.20	33,608
Share aged 65 and older (%)	17.5	8.03	34,952
Population density (residents/km^2)	4,910	6,010	36,013
Urban-area indicator (%)	51.1	50.0	46,544
Panel B. Road Environmental Census			
Land prices			
Land price (¥1,000/m^2)	135	171	68,288
Noise exposure			
24-hour noise level (dB)	68.2	5.45	19,048
Noise-control infrastructure			
Low-noise pavement coverage (%)	41.7	49.1	19,048
Noise-barrier coverage (%)	10.8	29.8	19,048
Roadside buffer-strip coverage (%)	2.99	16.4	19,048
Sound-absorbing panel coverage (%)	0.883	9.05	19,048
Traffic conditions			
Traffic volume (vehicles/12 hours)	23,200	13,400	17,606
Travel speed (km/h)	32.0	14.0	17,606
Heavy-vehicle share (%)	19.4	9.24	17,606

Notes: Panel A covers 2002–2022 and includes demographic information from the 2000 Population Census. College graduates include university and graduate-school graduates. Panel B uses the 2000–2010 Road Environmental Census releases, linked 1999 Road Traffic Census variables, and 2000–2010 Public Notice of Land Prices observations. Land prices pool the DiD samples within 1,000 m, comprising 6,208 sites. Infrastructure statistics summarize segment-level coverage, with installed-length ratios capped at 100%. SD denotes standard deviation; N counts nonmissing observations, so regression samples may be smaller.

B Trends

B-1 Sensitivity Analysis

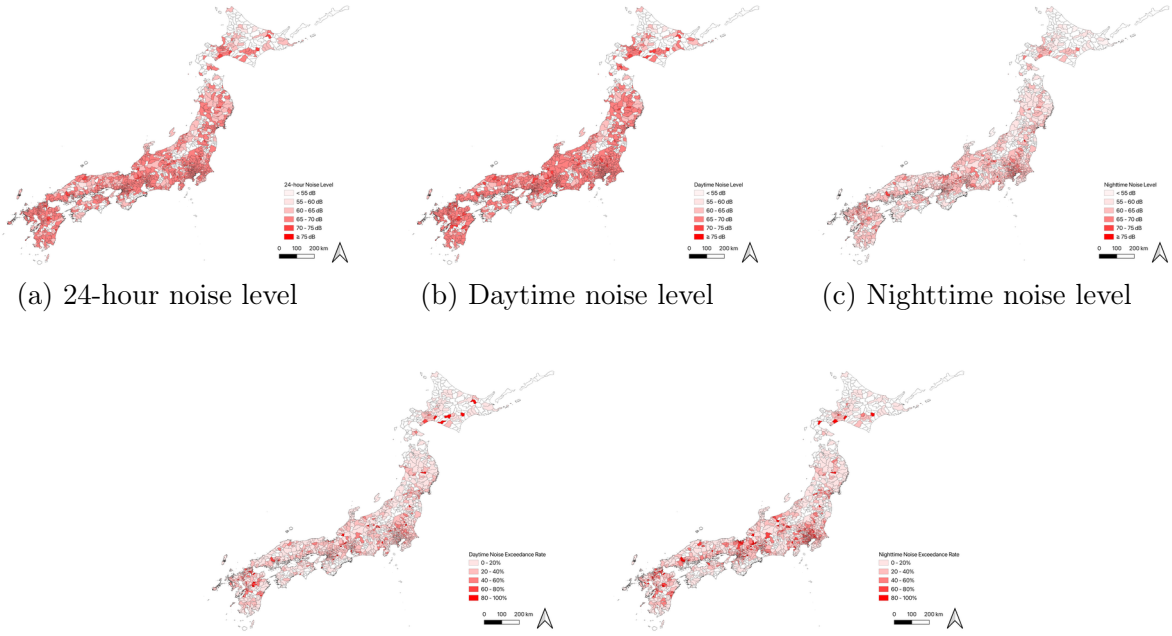
We assess the sensitivity of the parametric trend estimates in equation 3 to alternative specifications and samples (Appendix Table B1). Panel A reports specifications with a linear year trend. Panel B reports specifications with a quadratic year trend, using report year minus 2002 and its square. Columns (1) through (3) report trends in 24-hour, daytime, and nighttime noise levels. Columns (4) and (5) report trends in the share of observations exceeding the applicable daytime and nighttime environmental standards.

We vary outcome definitions and weighting rules to assess whether these choices affect the estimated trends. Panels A0 and B0 restate the main linear and quadratic estimates. Panels A1–A2 and B1–B2 use acoustic-energy outcomes and standardized noise outcomes. The main specifications are unweighted, so they estimate trends for an average monitored road segment. Panels A3 and B3 instead weight observations by evaluation-section length, estimating trends for an average location along monitored roads.

We also restrict the sample to assess sensitivity to monitoring frequency, expanding coverage, and the COVID-19 years (Panels A4–A6 and B4–B6). The Ministry of the Environment recommends measuring road-traffic noise at roughly five-year intervals, but in practice monitoring frequency varies across road segments and municipalities. Panels A4 and B4 therefore restrict the sample to road segments observed at least four times. Because monitoring coverage expanded over the sample period, Panels A5 and B5 restrict the sample to road segments observed in 2002–2006. Finally, Panels A6 and B6 exclude the COVID-19 years 2020–2021, when travel behavior and traffic volume may have been unusual.

Most specifications show declines in road-traffic noise over the sample period, with magnitudes close to the main estimates. In Panel A, the estimated annual decline in 24-hour noise is between 0.122 and 0.135 dB across the main level specification, the road-length-weighted specification, and the sample restrictions, compared with 0.127 dB in Panel A0. Daytime and nighttime levels show similar stability, and the exceedance-rate estimates remain negative in every specification. Panel B shows a negative linear term and positive quadratic term across specifications, indicating that noise declined more rapidly early in the sample and more gradually later on.

Figure B1: Road traffic noise levels and environmental-standard exceedance, 2002–2022

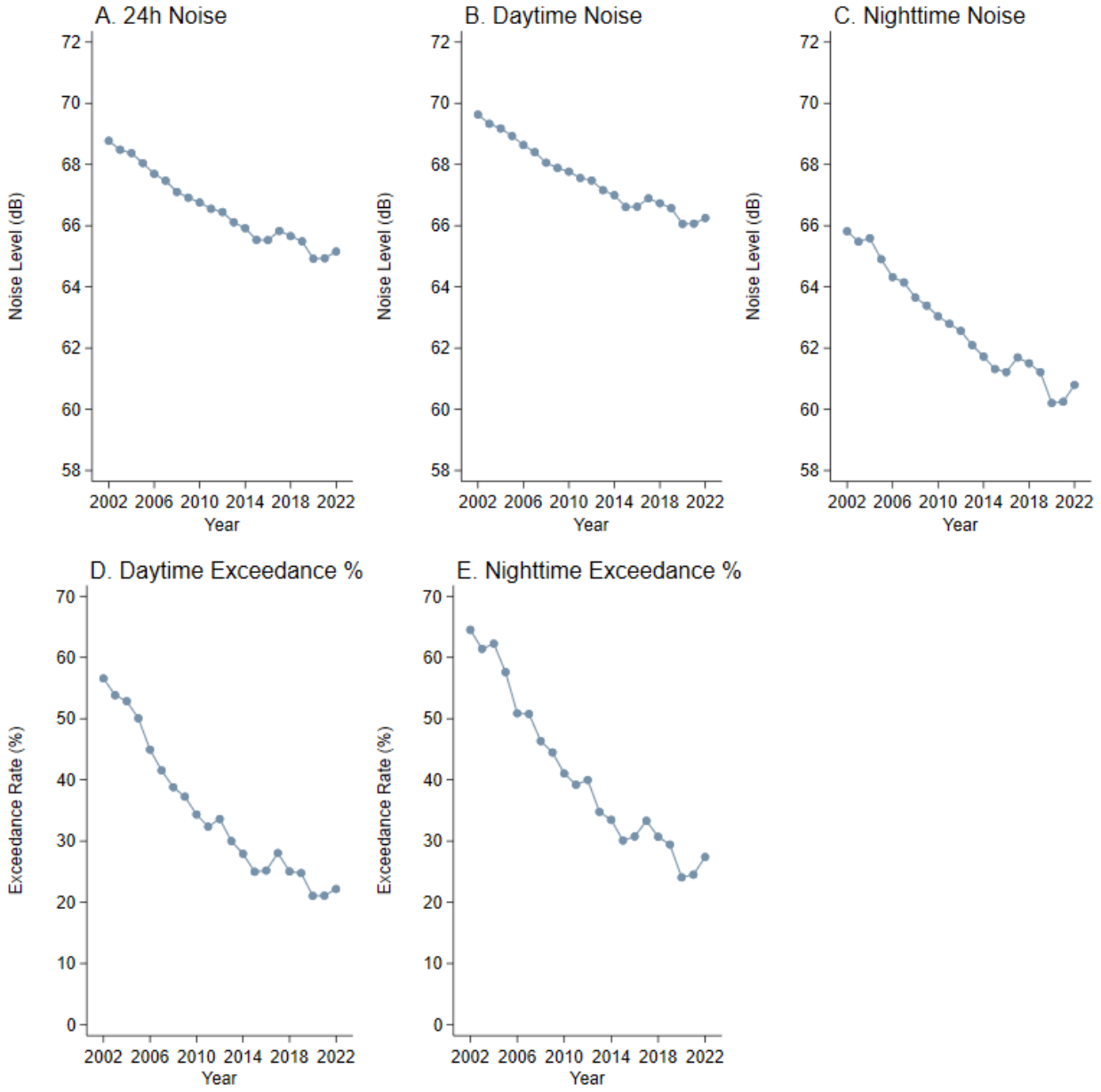


(a) 24-hour noise level (b) Daytime noise level (c) Nighttime noise level

(d) Daytime exceedance rate (e) Nighttime exceedance rate

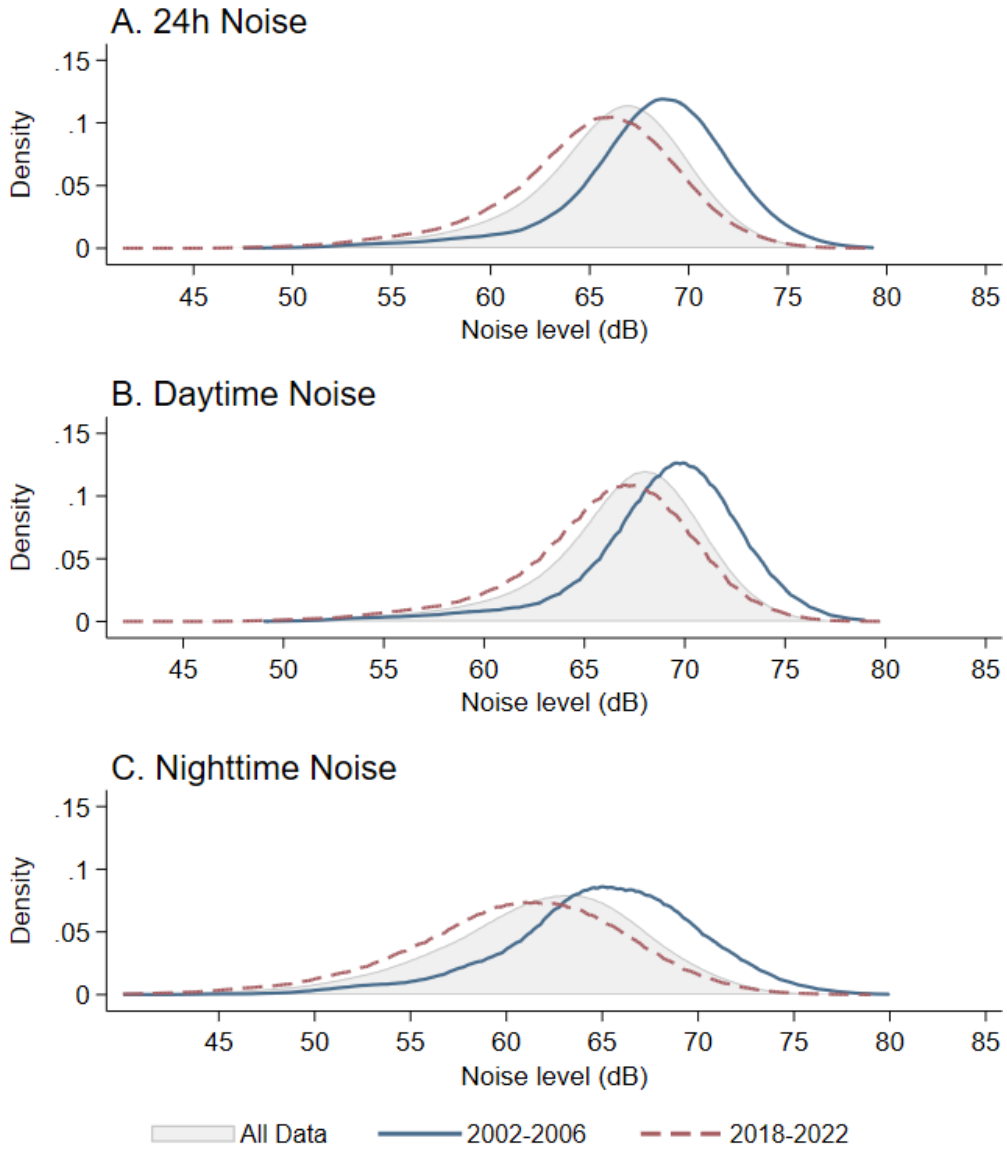
Notes: Panels A–C map municipality-level average 24-hour equivalent, daytime, and nighttime road traffic noise levels, respectively. The 24-hour equivalent noise level is constructed from daytime and nighttime equivalent noise levels. Panels D and E map municipality-level exceedance rates for daytime and nighttime environmental standards. The daytime exceedance rate is the share of monitoring observations with daytime noise exceeding 70 dB, and the nighttime exceedance rate is the share with nighttime noise exceeding 65 dB. White areas indicate municipalities with no matched monitoring observations during the sample period.

Figure B2: Trends in Average Road Traffic Noise and Exceedance Rates, 2002–2022



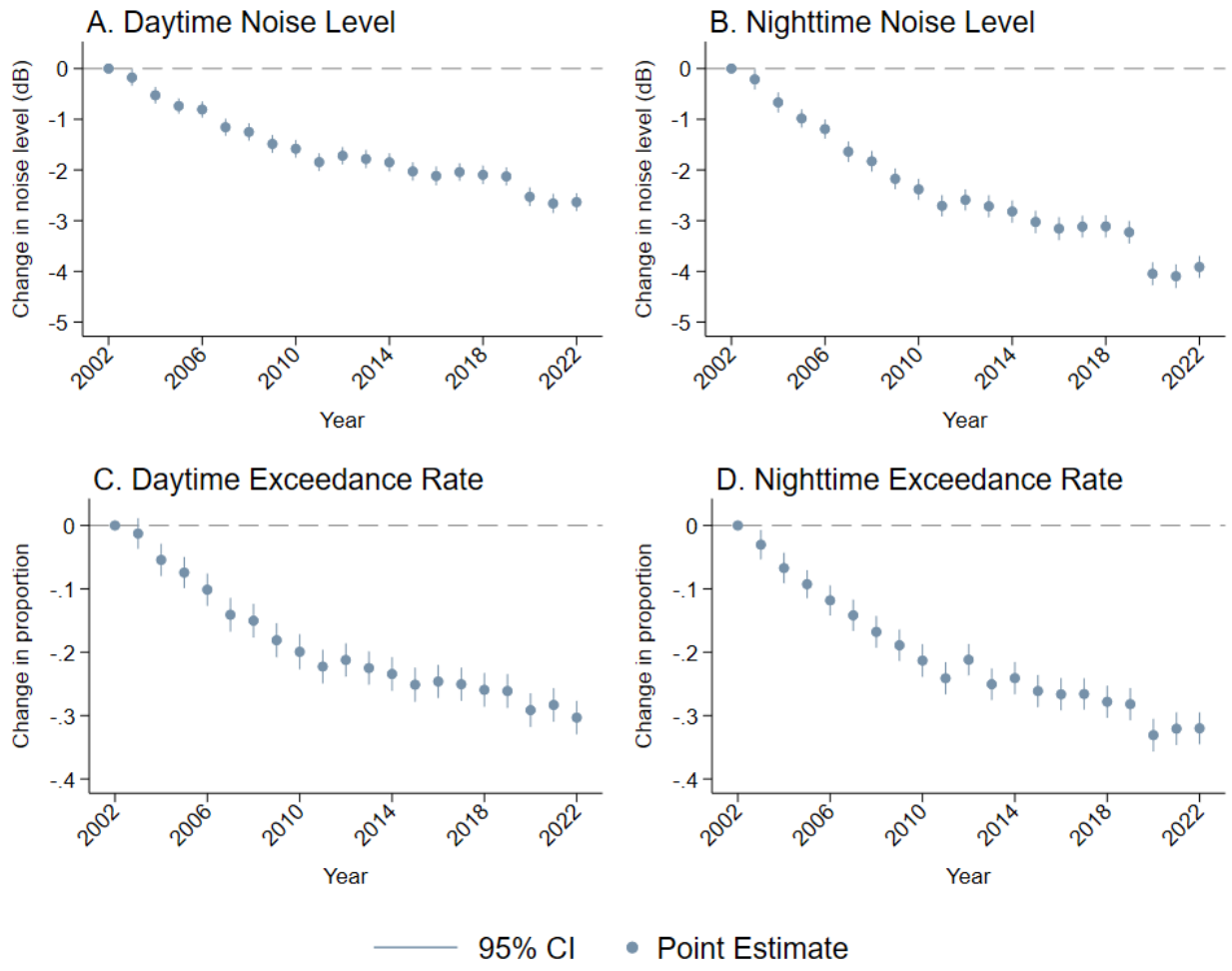
Notes: This figure plots annual average road traffic noise outcomes from 2002 to 2022. Panel A reports the 24-hour equivalent continuous sound level; Panel B reports daytime noise levels; Panel C reports nighttime noise levels; Panel D reports the daytime environmental-standard exceedance rate; and Panel E reports the nighttime environmental-standard exceedance rate. Each series is constructed by averaging monitoring-point-level observations within each report year. Exceedance rates are reported in percentage points.

Figure B3: Distribution of Road Traffic Noise Levels



Notes: This figure plots kernel density distributions of road traffic noise levels. Panel A reports the 24-hour equivalent continuous sound level; Panel B reports daytime noise levels; and Panel C reports nighttime noise levels. The figure compares distributions for the early period, 2002–2006, and the late period, 2018–2022. The shaded distribution shows the distribution based on all available years. Noise levels are measured in decibels.

Figure B4: Trends in Daytime and Nighttime Road Traffic Noise, 2002–2022



Notes: This figure plots estimated year-specific coefficients for daytime and nighttime road traffic noise outcomes. The omitted reference year is 2002. Panels A and B report changes in daytime and nighttime noise levels, respectively, measured in decibels. Panels C and D report changes in the proportions of monitoring points exceeding daytime and nighttime noise standards, respectively. The specification includes all controls and road-segment fixed effects. Standard errors are clustered at the road-segment level. Vertical bars represent 95 percent confidence intervals.

Table B1: Robustness of Estimated Trends in Road Traffic Noise

Specification	Trend	Noise level			Environmental standard exceedance rate	
		24h	Daytime	Nighttime	Daytime	Nighttime
		(1)	(2)	(3)	(4)	(5)
Panel A. Linear trend						
A0. Main	Year	-0.127***	-0.116***	-0.182***	-0.013***	-0.014***
		(0.003)	(0.003)	(0.004)	(0.000)	(0.000)
A1. Energy outcome	Year	-0.021***	-0.019***	-0.026***	-0.013***	-0.014***
		(0.001)	(0.001)	(0.001)	(0.000)	(0.000)
A2. Z-score outcome	Year	-0.028***	-0.027***	-0.031***	-0.013***	-0.014***
		(0.001)	(0.001)	(0.001)	(0.000)	(0.000)
A3. Road-length weighted	Year	-0.122***	-0.112***	-0.177***	-0.014***	-0.014***
		(0.004)	(0.004)	(0.005)	(0.001)	(0.001)
A4. At least 4 observations	Year	-0.129***	-0.118***	-0.186***	-0.014***	-0.015***
		(0.004)	(0.004)	(0.004)	(0.001)	(0.001)
A5. Observed in 2002-2006	Year	-0.135***	-0.123***	-0.194***	-0.015***	-0.016***
		(0.004)	(0.004)	(0.005)	(0.001)	(0.001)
A6. Excluding 2020-2021	Year	-0.127***	-0.117***	-0.178***	-0.014***	-0.015***
		(0.004)	(0.004)	(0.004)	(0.001)	(0.000)
Panel B. Quadratic trend						
B0. Main	Year	-0.223***	-0.202***	-0.307***	-0.028***	-0.028***
		(0.010)	(0.010)	(0.012)	(0.001)	(0.001)
	$Year^2$	0.005***	0.004***	0.006***	0.001***	0.001***
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
B1. Energy outcome	Year	-0.049***	-0.044***	-0.068***	-0.028***	-0.028***
		(0.002)	(0.002)	(0.003)	(0.001)	(0.001)
	$Year^2$	0.001***	0.001***	0.002***	0.001***	0.001***
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
B2. Z-score outcome	Year	-0.050***	-0.046***	-0.053***	-0.028***	-0.028***
		(0.002)	(0.002)	(0.002)	(0.001)	(0.001)
	$Year^2$	0.001***	0.001***	0.001***	0.001***	0.001***
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
B3. Road-length weighted	Year	-0.210***	-0.190***	-0.287***	-0.028***	-0.027***
		(0.014)	(0.013)	(0.016)	(0.002)	(0.002)
	$Year^2$	0.004***	0.004***	0.005***	0.001***	0.001***
		(0.001)	(0.001)	(0.001)	(0.000)	(0.000)

Continued on next page

Table B1 continued

Specification	Trend	Noise level			Environmental standard exceedance rate	
		24h	Daytime	Nighttime	Daytime	Nighttime
		(1)	(2)	(3)	(4)	(5)
B4. At least 4 observations	Year	-0.237***	-0.216***	-0.323***	-0.029***	-0.028***
		(0.012)	(0.012)	(0.014)	(0.002)	(0.002)
	$Year^2$	0.005***	0.005***	0.007***	0.001***	0.001***
		(0.001)	(0.001)	(0.001)	(0.000)	(0.000)
B5. Observed in 2002-2006	Year	-0.247***	-0.226***	-0.332***	-0.029***	-0.028***
		(0.012)	(0.012)	(0.014)	(0.002)	(0.002)
	$Year^2$	0.006***	0.005***	0.007***	0.001***	0.001***
		(0.001)	(0.001)	(0.001)	(0.000)	(0.000)
B6. Excluding 2020-2021	Year	-0.235***	-0.213***	-0.330***	-0.029***	-0.028***
		(0.011)	(0.010)	(0.012)	(0.002)	(0.001)
	$Year^2$	0.006***	0.005***	0.008***	0.001***	0.001***
		(0.000)	(0.000)	(0.001)	(0.000)	(0.000)

Notes: This table reports robustness checks for estimated time trends in road traffic noise. Panel A reports estimates from specifications with a linear year trend. Panel B reports estimates from specifications with a quadratic year trend, using the centered year variable and its square, where the centered year equals the report year minus 2002. Columns (1)–(3) report noise levels, and Columns (4)–(5) report the share of observations exceeding the applicable environmental standard. The energy outcomes are normalized by the baseline-year mean of the corresponding sound-energy measure. The z-score outcomes are standardized using the full sample. The road-length weighted specification uses positive, nonmissing evaluation-section length as an analytical weight. Each specification uses its own available sample. All specifications include the full set of controls and road-segment fixed effects. Standard errors clustered at the road-segment level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C Causes of Noise Reduction

C-1 Construction of Model Inputs

We construct the inputs used in the engineering model for each year from 2002 to 2022. Some data cover only selected years or end before 2022, so we interpolate or extrapolate where necessary. Traffic volume and the hybrid share use national series. For the remaining inputs, we estimate each prefecture's annual values and then aggregate them using noise-monitoring counts.

C-1.1 Traffic Conditions

We use the 2005, 2010, 2015, and 2021 Road Traffic Census to estimate traffic speed and heavy-vehicle shares in each prefecture (MLIT, 2021b). We interpolate between census years and extrapolate before the first census and after the last. We then aggregate the prefecture estimates to obtain national averages for traffic speed (v_t) and the heavy-vehicle share (ω_t) in each year (Panel B of Figure C1).

For traffic volume, we use national vehicle-kilometers from the Motor Vehicle Fuel Consumption Survey for 2010–2022 and the Motor Vehicle Transport Survey for 2002–2009, linked using MLIT's official vehicle-class conversion factors (MLIT, 2009b, 2022, nd). We divide each fiscal year's total by its number of days and express daily vehicle-kilometers as an index with 2002 equal to 100. This series supplies Q_t directly; we use no Road Traffic Census counts or calibration for volume. Figure C1 indexes all three traffic inputs to 100 in 2002 to show their relative changes.

We use national registered-vehicle distance as a proxy for traffic on monitored roads. Fiscal-year labels provisionally match the years in the noise data. The 2010 and 2011 totals include MLIT's estimates for observations missing after the earthquake, and the source of commercial-bus mileage changes in 2013. The fuel survey covers gasoline, diesel, LPG, and CNG vehicles; it excludes battery-electric vehicles and motorcycles. These coverage and timing differences limit how closely the national series can track traffic at individual monitoring sites (MLIT, nd).

C-1.2 HV/PHV Share

We calculate the HV/PHV share (h_t) using observed national vehicle-stock data from the Automobile Inspection and Registration Information Association (AIRIA) (AIRIA, 2025a,b). For each year, we divide the number of hybrid and plug-in hybrid passenger vehicles by the total passenger-vehicle stock. These data cover every year from 2002 to 2022, so we need no interpolation or extrapolation.

C-1.3 Noise Emission Standards

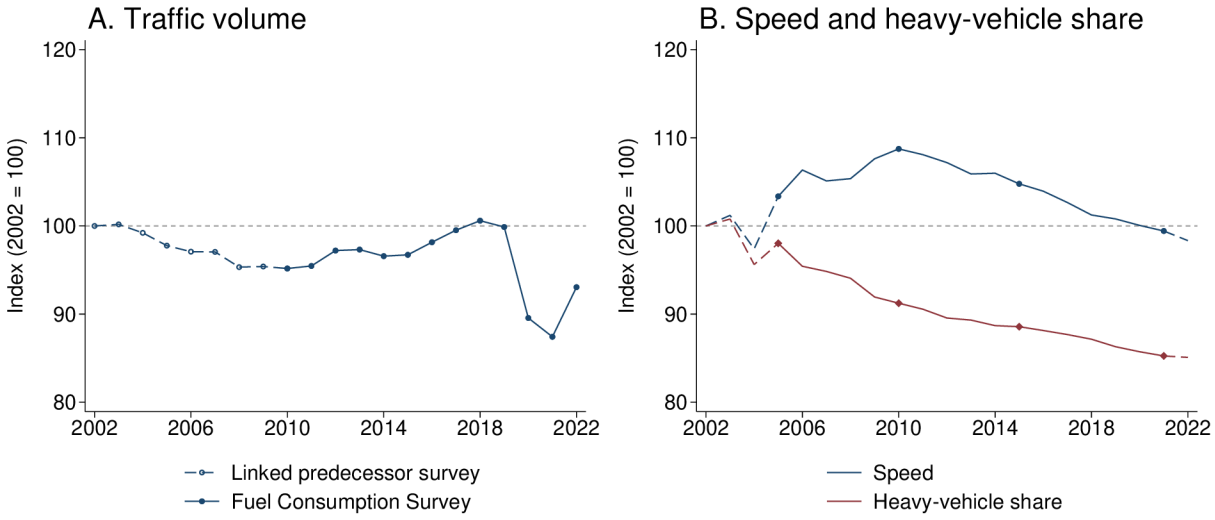
We estimate the shares of light and heavy vehicles that meet the tighter standards (s_t^S and s_t^L) separately for each prefecture, using vehicle sales and stock data and assumptions about fleet replacement. First, we divide annual new light-vehicle sales by the existing light-vehicle stock and use this ratio to approximate the replacement rate for both groups (Zenkeijikyo, 2025). Second, we assume that vehicle owners replace vehicles that predate the tighter standards at three times the rate of newer vehicles. We multiply the cumulative replacement share by 95%, so coverage cannot exceed 95%. We track replacement from 1999 onward for light vehicles and from 2002 onward for heavy vehicles. We then aggregate the prefectural estimates into national shares.

C-1.4 Road Infrastructure

We construct road infrastructure inputs from the Road Environmental Census for 2000–2010: coverage by low-noise pavement (c_t^p), noise barriers (c_t^b), roadside buffer strips (c_t^e), and sound-absorbing panels (c_t^A). To our knowledge, no comprehensive statistics track the coverage of these measures since the Road Environmental Census ended. We fit an exponential saturation curve for each prefecture and measure using its 2000–2010 trend, then project coverage through 2022. These curves allow expansion to slow as coverage approaches a ceiling, with the expansion rate varying across prefectures. We use the 90th percentile of coverage across prefectures in 2010 as the ceiling—for example, 78% for pavement and 22% for noise barriers. If a prefecture already exceeds this level, we use its own 2010 coverage as its ceiling. We then aggregate the prefecture estimates to obtain annual national coverage (Figure C2).

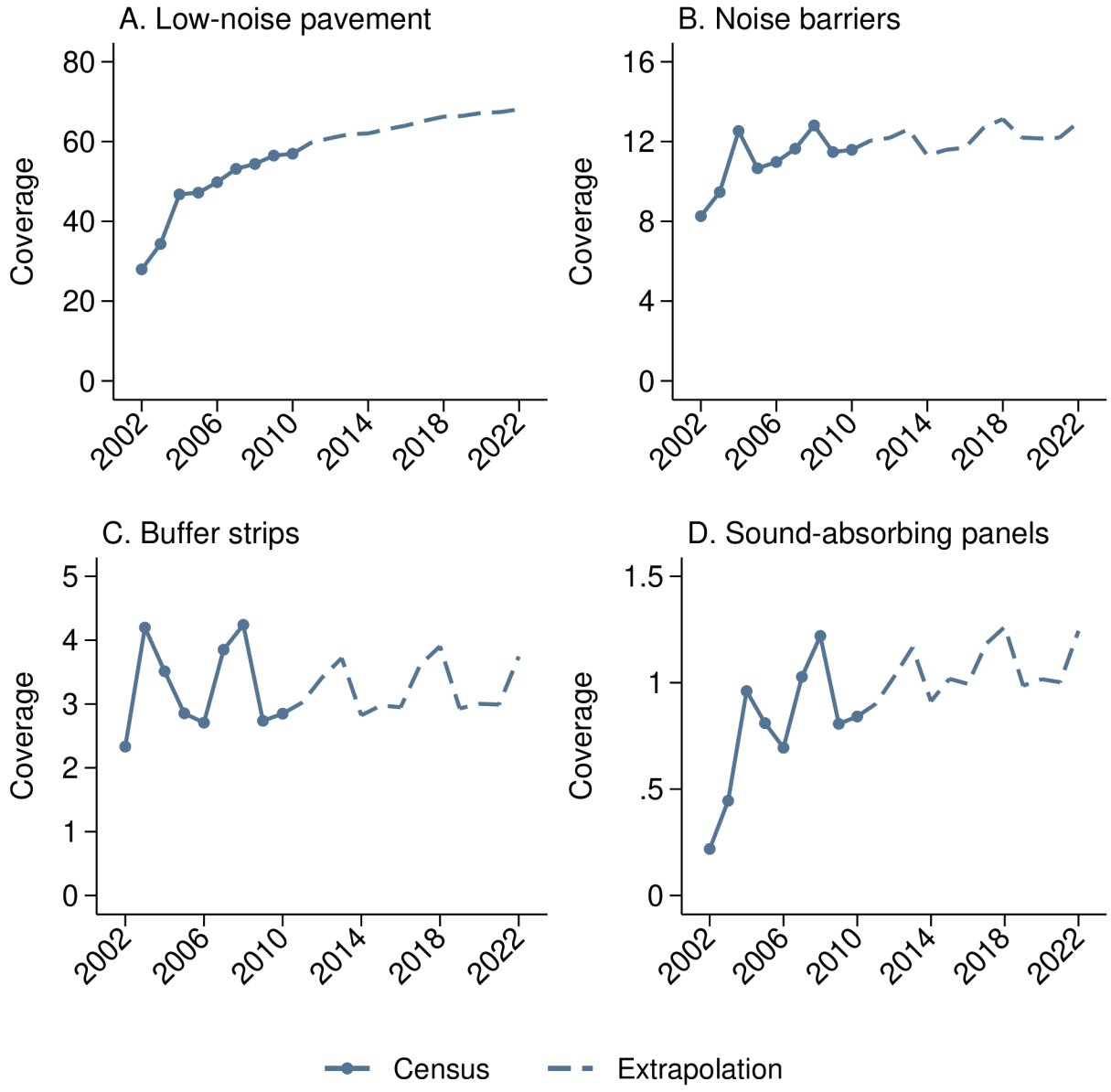
C-2 Figures and Tables

Figure C1: Traffic Volume, Speed, and Heavy-Vehicle Share



Notes: Each series equals 100 in 2002, and both panels use the same vertical scale. A value of 90 indicates a 10% decline from 2002. Panel A shows national daily vehicle-kilometers. The dashed segment uses the officially linked Motor Vehicle Transport Survey before 2010; the solid segment uses the Fuel Consumption Survey from 2010. Panel B shows indices of speed and the heavy-vehicle share. Markers show census observations, solid lines show estimates between census years, and dashed lines extend beyond them.

Figure C2: Road Infrastructure Inputs: Census Trends and Extrapolation



Notes: Coverage shows the percentage of road length with each measure. Solid lines show the census series; dashed lines show projections after 2010. Each panel uses a different vertical scale.

Table C1: Inputs Used in the Engineering Predictions

Variable	2002–2012			2013–2022		
	2002 (1)	2012 (2)	Change (3)	2013 (4)	2022 (5)	Change (6)
Panel A. Traffic conditions						
Traffic volume (2002 = 100)	100.0	97.2	−2.8%	97.3	93.1	−4.4%
Speed (km/h)	32.8	35.2	+7.2%	34.8	32.3	−7.1%
Heavy-vehicle share	14%	13%	−1.5 ppt	13%	12%	−0.6 ppt
Panel B. HV/PHV adoption						
HV/PHV share of passenger vehicles	0.14%	3.4%	+3.3 ppt	4.8%	17%	+13 ppt
Panel C. Vehicle noise standards						
Light vehicles covered by tighter standards	36%	81%	+45 ppt	83%	92%	+9 ppt
Heavy vehicles covered by tighter standards	8.8%	66%	+58 ppt	70%	88%	+18 ppt
Panel D. Road infrastructure						
Low-noise pavement coverage	28%	61%	+33 ppt	62%	68%	+6.3 ppt
Noise-barrier coverage	8.3%	12%	+3.9 ppt	13%	13%	+0.35 ppt
Sound-absorber coverage	0.22%	1%	+0.81 ppt	1.2%	1.2%	+0.078 ppt
Buffer-zone coverage	2.3%	3.4%	+1.1 ppt	3.7%	3.7%	+0.024 ppt

Notes: Traffic volume is a national daily vehicle-kilometer index (2002 = 100). Changes in traffic volume and speed use percent (%); changes in shares use percentage points (ppt). We calculate changes before rounding. HV and PHV denote conventional hybrids and plug-in hybrids.

Table C2: Model Fit Against Year-Fixed-Effect Benchmarks

Statistic	2002–2012 (1)	2013–2022 (2)	2002–2022 (3)
Bias (dB)	−0.02	0.05	0.02
MAE (dB)	0.20	0.11	0.16
RMSE (dB)	0.26	0.14	0.21
Corr.	0.93	0.97	0.97

Notes: Prediction error equals the model's noise change since 2002 minus the regression estimate. Bias shows the average error. MAE shows the average size of the error, ignoring its sign; RMSE gives more weight to larger errors. Corr. shows how closely the two series move together.

Table C3: Parameter Ranges Used in the Noise Decomposition

Mechanism	Parameter (1)	Lower value (2)	Higher value (3)	Interpretation (4)
Panel A. Traffic conditions				
Traffic volume	α_Q	0.8	1.2	Acoustic-energy elasticity with respect to traffic volume
Traffic speed	γ	0.8	1.2	Acoustic-energy elasticity with respect to speed
Large-vehicle share	δ_L	6.0 dB	8.5 dB	Large-vehicle noise premium relative to small vehicles
Panel B. HV/PHV adoption				
HV/PHV vehicles	d_H	1.0 dB	2.0 dB	Noise reduction relative to comparable conventional vehicles
Panel C. Vehicle noise standards				
Small vehicles	d_s	1.8 dB	2.6 dB	Tightening of light-vehicle acceleration-noise limits
Large vehicles	d_L	2.2 dB	3.0 dB	Tightening of heavy-vehicle acceleration-noise limits
Road-level pass-through	θ_R	0.4	0.6	Assumed multiplier on the single-vehicle dB reduction
Panel D. Road infrastructure				
Drainage pavement	d_P	2.0 dB	3.0 dB	Expected noise reduction
Environmental facility belt	d_E	3.0 dB	8.0 dB	Expected noise reduction
Noise barrier	d_B	3.0 dB	8.0 dB	Expected noise reduction
Sound-absorbing panel	d_A	1.5 dB	5.0 dB	Expected noise reduction

Notes: We vary the parameters within these ranges to check how the results change. Table 3 reports the baseline values.

Table C4: Robustness of the Noise Decomposition to Parameter Assumptions

Decomposition term	Reduction magnitude (dB)	
	Minimum (1)	Maximum (2)
Panel A. Traffic conditions	0.51	0.79
Traffic volume	0.25	0.38
Speed	0.06	0.09
Heavy-vehicle share	0.20	0.32
Panel B. HV/PHV adoption	0.09	0.15
Panel C. Vehicle noise standards	0.52	1.18
Panel D. Road infrastructure	1.00	1.74
Pavement	0.80	1.20
Noise barriers	0.14	0.38
Other road infrastructure	0.06	0.16
Predicted reduction	2.13	3.85
Year-FE decline	2.84	2.84

Notes: Values show the smallest and largest noise reductions from 2002 to 2022 across the parameter choices in Table C3. Each row can reach its limits under different choices, so the rows need not add to the total.

D Effects of Pavement Conversion

D-1 Additional Results

Table D1: Effects of Pavement Conversion on Air Pollution

	Dependent variable		
	NOx (ppb)	SPM (μg/m^3)	CO (ppb)
Conversion effect	1.26	-1.20	25.17
	(1.89)	(0.93)	(26.71)
Treated monitors	55	51	32
Pre-treatment mean	65.2	35.0	732.2
Road structure controls	Yes	Yes	Yes
Observations	2,431	2,285	1,469

Notes: Callaway–Sant'Anna, never-treated comparison, monitor and year fixed effects, clustered by segment. Roadside and general-environment monitors matched to a census segment within 100m; a monitor is treated when its segment converted from dense-graded to porous asphalt, and monitors matched to conversions from other surfaces are dropped. Controls are the matched segment's road width and number of lanes at its first observation. NOx and CO are in ppb and SPM in μg/m^3; the pre-treatment mean is the level in the treated monitors before conversion. No pollutant shows a significant effect. The sample is 2000–2010, as elsewhere in the paper; where a monitor matches more than one census segment the nearest is kept. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D-2 Details on Benefits and Costs Calculations

D-2.1 Overview

This appendix section elaborates on the benefit and cost calculations in Section 6.5. We compute the benefit and the cost separately for each converted segment and report medians across segments in Table D2. The sample is the 1,096 segments converted from dense-graded to porous asphalt between 2000 and 2010, which together cover 1,935 km of road.

We estimate the benefits of low-noise pavement by applying the estimated 3.6% increase in land prices to the baseline value of residential land within 100m of each road segment. Table D2 summarizes the calculation. We measure the residential land within 100m of each road segment from the 100m land-use mesh and value it at the pre-conversion land price per square meter. Across converted segments, the median residential land value is 3.9 billion yen, implying a median benefit of 140 million yen (US$1.2 million). [2]

In contrast, the cost of low-noise pavement includes both initial conversion and subsequent resurfacing. In principle, road authorities in Japan inspect all roads every five years and resurface them every ten years. The cost of installing low-noise pavement is therefore the additional expense of resurfacing with low-noise porous asphalt rather than conventional dense-graded asphalt. We assume that households expect current noise conditions to persist when purchasing a home. Thus, we include the additional cost of maintaining low-noise pavement through resurfacing every ten years, alongside the initial conversion cost. We discount the additional costs of future resurfacing at 4%. Together, these costs have a median present value of 27 million yen (US$0.23 million).

[2] We use an exchange rate of 120 yen per U.S. dollar throughout this paper.

Table D2: Cost-Benefit Analysis of Pavement Conversion from Dense to Porous Asphalt

	Median across converted segments
Panel A: Benefits	
Segment length	1.3 km
Residential land within 100m	4.2 ha
Pre-conversion land price	93,700 yen/m^2
Residential land value	3.9 billion yen
Capitalized benefit	140 million yen
Panel B: Costs	
Paved area	30,900 m^2
Cost difference, porous − dense (discounted)	27 million yen
Panel C: Benefits against costs	
Benefit/cost ratio	4.99
charging the full porous resurfacing	0.97

Notes: Medians across the 1,096 segments converted from dense-graded to porous asphalt, each row computed separately, so the column does not describe a single segment. Each segment's residential land is the built-up land in residential zones (MLIT land-use mesh and zoning) within 100m of the road; land within 100m of two converted segments is assigned to the nearer one. Land is valued at the Official Land Price of the nearest surveyed site in the year before conversion; the price row covers segments with residential land. The benefit applies the triple-difference estimate within 100m (3.6 percent; Table 5, column (2) of Panel B). The cost difference is the premium of porous over dense-graded asphalt for the segment's lane count (MLIT standard unit prices for 2025, deflated to 2000–2010), applied to its paved area at conversion and every 10 years and discounted at 4 percent. Benefit-cost ratios are computed for each segment before taking the median; the full-resurfacing row charges the entire porous surface course.

D-2.2 Detailed Calculations

Residential Land. We intersect a 100m band on either side of each converted road with the 100m land-use mesh (MLIT, 2009a) and keep the built-up land in residential zones (MLIT, 2011). Where the bands of two converted segments overlap, we assign the land to the nearer segment, so each cell counts once. The median segment is 1.3 km long and has 4.2 hectares of residential land within 100m. About 28 percent of segments have no residential land within 100m, so their benefit is zero.

The Price of Land. We value each residential cell at the price of the nearest site in the Official Land Prices in the year before the segment's conversion. Summing area times price over a segment's cells gives the value of its residential land, which is 3.9 billion yen at the median. The benefit applies the 3.6 percent price effect within 100m (Table 5) to this value, which gives a median benefit of 140 million yen.

The Cost of Porous Surfacing. As we state in Section 6.5, we treat the cost of conversion as the extra cost of resurfacing with porous rather than dense-graded asphalt. We take both prices from MLIT's standard unit prices for 2025 (MLIT, 2025b) and deflate them to the 2000 to 2010 average with the road-pavement construction-cost deflator (MLIT, 2026). The schedule prices porous surfacing by lane count, so each segment receives the premium for its own lane count, 358 yen per square meter for one lane per direction and 251 yen for two or more. The schedule also prices porous surfacing with and without a drainage pipe, which the data cannot settle because the panel records no kerb or gutter. We assume pipes throughout, which is standard on a kerbed carriageway and is the conservative choice.

Paved Area and Lifecycle Cost. Each segment's paved area is its length times its road width, which is 30,900 square meters at the median. We assume that road authorities resurface roads every 10 years regardless of the surface type, consistent with the design period for asphalt pavement (MLIT, 2001). The premium is therefore incurred at conversion and again at every resurfacing. We discount it continuously at 4 percent (MLIT, 2025a), so its present value is $1/(1-e^{-0.4}) \approx 3.03$ times a single premium. This gives a median cost of 27 million yen.

Benefit-Cost Ratios. We compute the ratio of benefit to cost for each segment and then take the median, which is 5.0. Because the medians in Table D2 are taken row by row, this ratio differs from the ratio of the median benefit to the median cost. Charging the full price of porous resurfacing at conversion and every ten years, rather than only the premium, gives a median ratio of 0.97. Among segments with residential land within 100m, this full-cost ratio is 2.0.

E Construction of the Welfare Quantities

This appendix describes how we build the residential land stock in Table 6. The stock rests on three measurements along the monitored road network. These are the residential land within 100m of the road, the households living on that land, and its price.

Roadside band. The Ministry of the Environment publishes the evaluation area of each monitored road segment as a strip about 50m wide on each side of the road. Monitoring rotates across segments over a roughly five-year cycle, so a single year covers only part of the network. We therefore take the union of the evaluation areas from every year between 2002 and 2023. We then widen this union so that it extends 100m from the road centerline, the distance over which we estimate the land-price effect in Section 6.3. The resulting band covers 15,319 km^2.

Residential land. We intersect the band with the 100m land-use mesh (MLIT, 2009a), which classifies 6,604 km^2 of the band as built-up land. We keep the built-up land that lies in one of the seven residential use districts of the zoning maps (MLIT, 2011). This leaves 2,465 km^2 of residential land.

Households. We take households and residents from the small-area tabulations of the 2010 Population Census (Statistics Bureau of Japan, 2010). Small areas are larger than the 100m mesh, so we assign each small area's households to the band in proportion to the share of its built-up land that lies inside the band. The band contains 15.5 million households and 36.2 million residents.

Land prices. We price each residential mesh cell at the nearest residential site in the Official Land Prices, using the site's mean price over 2002 to 2022. The band contains 1,832 such sites, and the median cell lies 0.8 km from its nearest site. The stock is the sum of area times price over all residential cells, which gives 222.5 trillion yen, or 90,267 yen per square meter on average. We weight prices by area rather than averaging the sites because posted prices are right-skewed and expensive sites cluster where the band is dense. Valuing all residential land at the median site price would raise the capitalized benefit from 6.8 trillion yen to 8.0 trillion, and at the mean site price to 12.9 trillion. Dropping the 8 percent of residential land that lies more than 5 km from a posted site lowers it to 6.5 trillion.

Capitalized benefit. The capitalized benefit multiplies the residential land stock by the willingness to pay per decibel (1.42 percent of land value) and by the decline attributed to policy (2.14 dB). The decline is the sum of the 2022 contributions of vehicle emission standards (0.79 dB), low-noise pavement (1.00 dB), and other road infrastructure (0.35 dB) in the engineering model of Section 5. We split the benefit across the three channels in proportion to these contributions. The per-household figure divides the benefit by the 15.5 million households in the band.

References for Appendix

AIRIA (2025a). Historical stocks of hybrid and electric vehicles. March-end stocks; the hybrid table includes plug-in hybrids. Available at https://www.airia.or.jp/publish/file/HVEV_2025.pdf.

AIRIA (2025b). Historical vehicle stocks by vehicle type. March-end stocks, including kei vehicles. Available at https://www.airia.or.jp/publish/file/hoyuudaisuusuii_2025.pdf.

EA (2000). Processing standards for continuous monitoring of automobile noise. Notification No. Kan-Dai-Ni 53, May 18. Available at https://www.env.go.jp/hourei/07/000015.html.

MLIT (2001). Technical standards for pavement structure. Notification No. Kokuto-Gai 48 and Kokudo-Ki 55, June 29. Fatigue-cracking requirements are stated per 10-year design period, and the design period is set by the road administrator. Japanese title: 舗装の構造に関する技術基準について.

MLIT (2002–2009b). Motor vehicle transport survey. Annual reports. Available at https://www.mlit.go.jp/k-toukei/jidousya202003.html.

MLIT (2005–2021b). Road traffic census: General traffic volume survey. Available at https://www.mlit.go.jp/statistics/details/sample03_2_00044.html.

MLIT (2009a). Land use subdivision mesh data. National Land Numerical Information, dataset L03-b. 100 m mesh (1/10 subdivision of the third-order mesh), FY2009 vintage. Japanese title: 土地利用細分メッシュデータ.

MLIT (2010–2022). Motor vehicle fuel consumption survey. Annual reports. Available at https://www.mlit.go.jp/k-toukei/nenryousyouhiryou.html.

MLIT (2011). Use district data. National Land Numerical Information, dataset A29. FY2011 vintage. Japanese title: 用途地域データ.

MLIT (2021a). Implementation of the 2021 National Road and Street Traffic Conditions Survey. https://www.mlit.go.jp/road/ir/ir-council/ict/pdf05/01.pdf. In Japanese; title translated.

MLIT (2025a). Cost-benefit analysis manual. Technical report, Road Bureau and City Bureau, Government of Japan. Sets the social discount rate at 4 percent.

MLIT (2025b). Standard unit prices for the construction-package costing method, 2025. Technical report, Government of Japan. Applies to works with a bid deadline between 1 April 2025 and 31 March 2026.

MLIT (2026). Construction cost deflator. Annual series, FY2020 = 100. Road pavement series; FY2000–2010 mean 79.1, 2025 provisional 121.8. Release of 30 June 2026.

MLIT (n.d.). Motor vehicle fuel consumption survey: Methods and notes for users. In Japanese. Definitions, survey coverage, links to the earlier Motor Vehicle Transport Survey, and the 2013 commercial-bus source change.

MOE (2005). Processing standards for continuous monitoring of automobile noise under article 18 of the noise regulation act. Notification No. Kan-Kan-Ji 050629002, June 29. Available at https://www.env.go.jp/hourei/07/000031.html.

MOE (2011). *Automobile Noise Continuous Monitoring Manual*. Available at https://www.env.go.jp/content/900398331.pdf.

MOE (2015). *Environmental Quality Standards for Noise Evaluation Manual: Areas Facing Roads*.

Available at https://www.env.go.jp/content/900400600.pdf.

NIES (2023). *Jidōsha Sōon no Jōji Kanshi Kekka: Daunrōdo Dēta Riyō Setsumeisho [User Guide for Downloaded Data on Continuous Monitoring of Road Traffic Noise]*. National Institute for Environmental Studies, Tsukuba, Japan. In Japanese.

NIES (2025). Environmental gis: Air-pollution monitoring data. Available at https://tenbou.nies.go.jp/gis/explain/atm/moni.html.

SBJ (2000). 2000 population census: Small-area tabulations. Tables on age composition and educational attainment. Available at https://www.stat.go.jp/data/kokusei/2000/kohyo.html.

Statistics Bureau of Japan (2010). 2010 population census of japan: Small-area tabulations and boundary data. e-Stat Statistical GIS, series 00200521. Households (`SETAI`) and population (`JINKO`) by small area (*chōchō*), with boundary geometry; 47 prefecture files.

Zenkeijikyo (2025). Kei-vehicle statistics. Available at https://www.zenkeijikyo.or.jp/statistics.